

Mobile Computing

Lecture 6 Bluetooth

د/ أحمد حسين
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Talk Outline

- Introduction to Bluetooth technology.
- Bluetooth features
- Bluetooth applications
- Bluetooth vs. 802.11b
- Bluetooth networking
 - Piconets
 - Scatternets
- Bluetooth Protocol Stack
- Bluetooth low-power states
- Bluetooth security



Introduction

- Bluetooth is an open wireless technology standard for exchanging data and voice over **short distances**.
- Bluetooth uses short wavelength radio transmissions in the Industrial, Scientific, and Medical (ISM) band from **2400-2480 MHz**.
- Bluetooth involves fixed and mobile devices, creating personal area networks (**PANs**) with high levels of security.
- The Bluetooth technology aims at so-called **ad-hoc piconets**, which are local area networks with a very limited coverage and without the need for an infrastructure.

Bluetooth

Features

- Bluetooth is wireless and **automatic**. The need to manage cables, connectors, and connections is eliminated, and initiating communications requires no special actions.
- Devices autonomously identify and initiate communication without user intervention, with the exception of situations where authentication is necessary, such as accessing email accounts that require login credentials.
- Bluetooth technology is characterized by its **affordability**. Market analysts estimate that the minimum cost of incorporating Bluetooth into a PDA, cell phone, or other product is relatively low.

Bluetooth

Features

- The ISM band utilized by Bluetooth is regulated but unlicensed. Governments have standardized this band, enabling the seamless use of the same devices across diverse locations without the need for prior legal authorization.
- Bluetooth seamlessly integrates **data and voice transmission capabilities**. This dual functionality enables innovative applications such as mobile hands-free headsets for voice communication, as well as applications that facilitate fax printing and synchronize address books across devices like PDAs, laptops, and cell phones.
- Signals are omnidirectional and can **traverse walls and briefcases**. Unlike infrared, communication devices do not require alignment or an unobstructed line of sight for effective communication.
- Bluetooth employs **frequency hopping**, a spread spectrum technique that significantly diminishes the likelihood of communications interception.

Bluetooth

Applications

- File Transfer.
- **Ad-hoc networking:** Communicating devices can spontaneously form a community of networks that persists only as long as it's needed.
- Device synchronization: **Seamless connectivity** among PDAs, computers, and mobile phones allows applications to update information on multiple devices automatically when data on any one device changes.
- Car kits: **Hands-free** packages enable users to access phones and other devices without taking their hands off the steering wheel.
- Bluetooth Low Energy (BLE) technology enables **seamless contactless transactions** between smartphones and payment terminals.

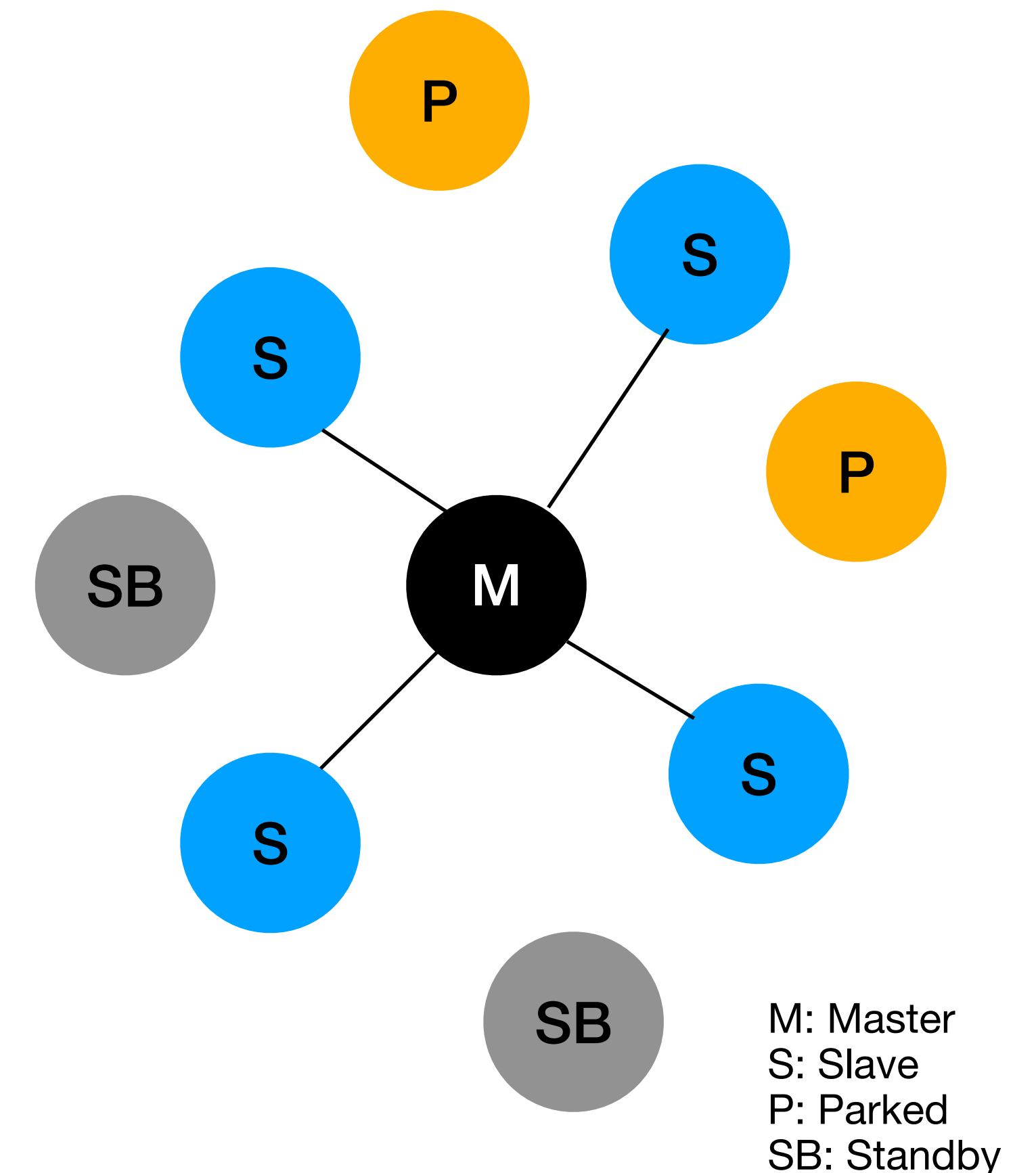
Bluetooth vs. 802.11b

- The 802.11b protocol is engineered to facilitate communication between substantial devices characterized by substantial power consumption and high data transfer rates, such as **desktops** and **laptops**. These devices engage in communication at a maximum speed of **11 Mbit/sec**, enabling them to connect over extended distances, reaching up to **100 meters**.
- In contrast, Bluetooth is engineered to facilitate connections between compact devices such as PDAs, mobile phones, and peripherals at reduced speeds (**1 Mbit/sec**) and within a confined range (**10 meters**). This design optimization minimizes power consumption.
- Another major difference is that 802.11b wasn't designed for voice communications, while any Bluetooth connection can support both **data and voice** communications.

Bluetooth

Networking

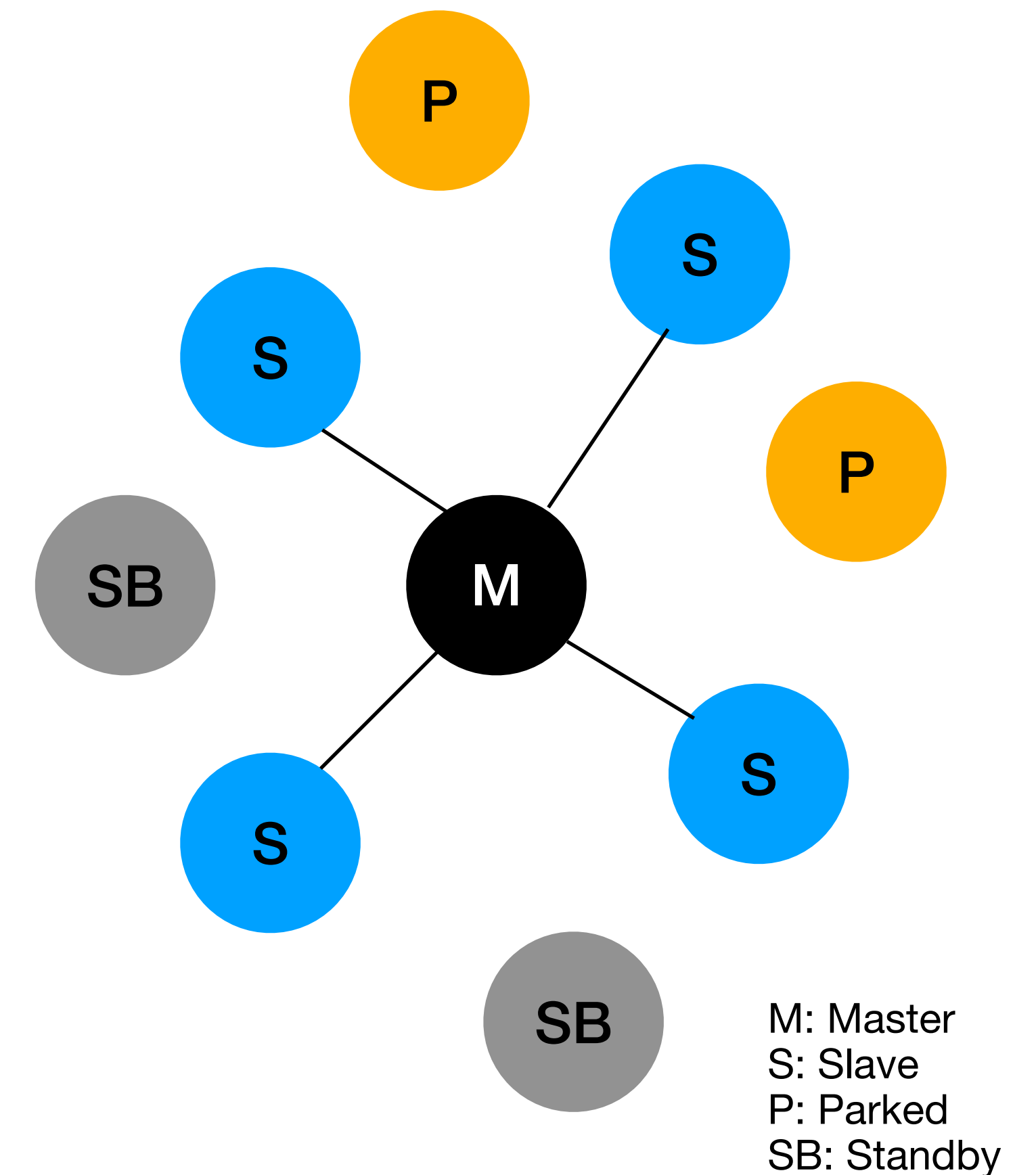
- Bluetooth operates on **79 channels** within the 2.4 GHz band, utilizing a 1 MHz carrier spacing.
- Each device employs frequency hopping with a rate of **1,600 hops per second** in a pseudorandom manner.
- A **piconet** is a collection of Bluetooth devices which are synchronized to the **same hopping sequence**.
- **One** device in the piconet can act as **master** (M), all other devices connected to the master must act as **slaves** (S).
- The **master** determines the **hopping pattern** in the piconet and the slaves have to synchronize to this pattern.



A Simple Piconet

Bluetooth Networking

- Each piconet has a unique **hopping pattern**.
- If a device wants to participate, it has to synchronize to this hopping pattern.
- **Parked devices** (P) are unable to actively participate in the piconet (i.e., they lack a connection). However, they are known and can be reactivated within a few milliseconds.
- More than 200 devices can be parked.
- Devices in **stand-by** (SB) do not participate in the piconet.
- Each piconet has **exactly one master** and up to **seven** simultaneous **slaves**.
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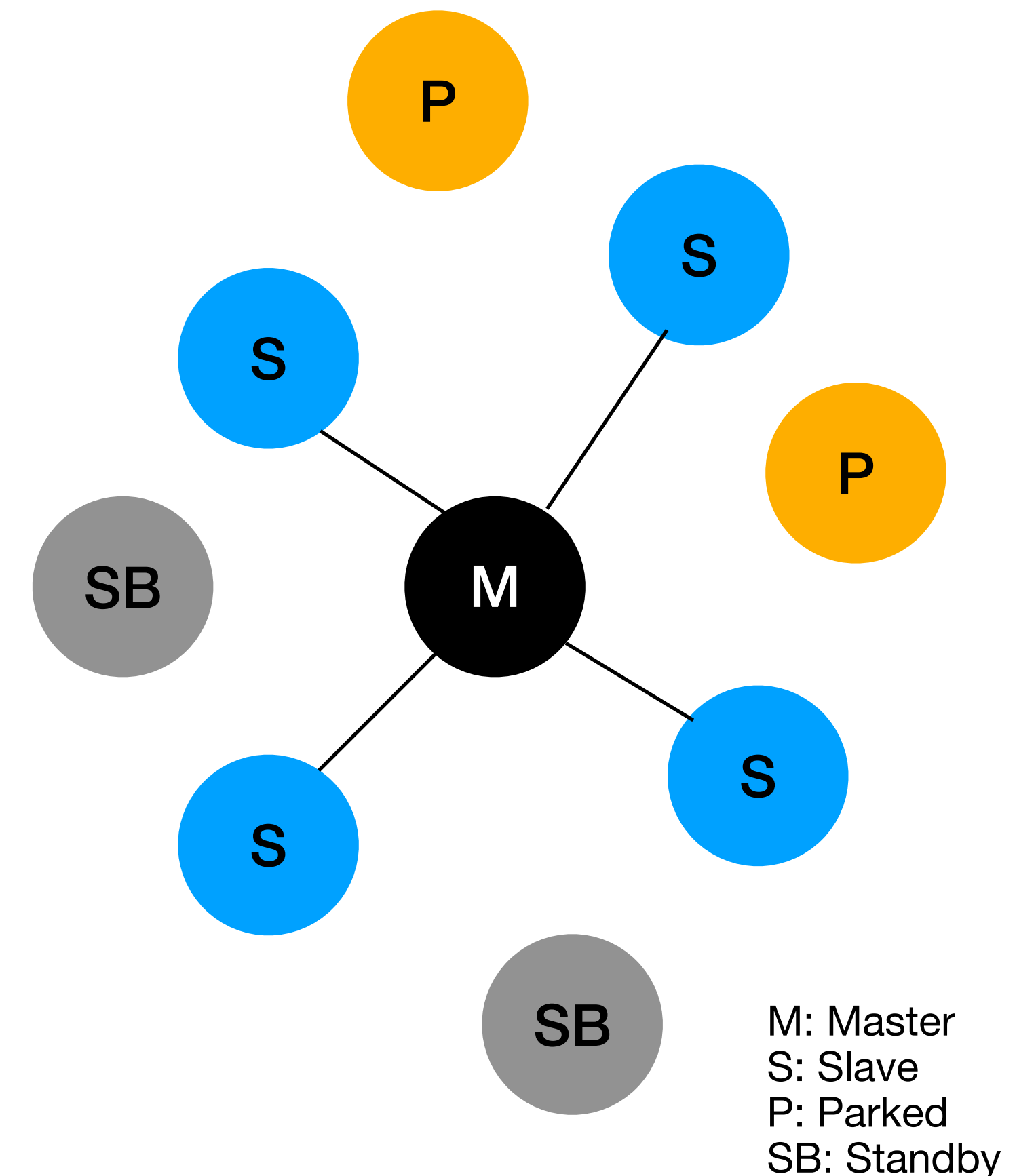


A Simple Piconet

Bluetooth

Forming a bluetooth piconet

- The initiation of a piconet process commences with a master transmitting its clock and device identification information.
- All the Bluetooth devices have the same capability to become a master or a slave and two or three devices are sufficient to form a piconet.
- The unit establishing the piconet automatically becomes the master, all other devices will be slaves.
- The *hopping pattern* is determined by the **device ID**, a 48-bit worldwide unique identifier.



A Simple Piconet

Bluetooth

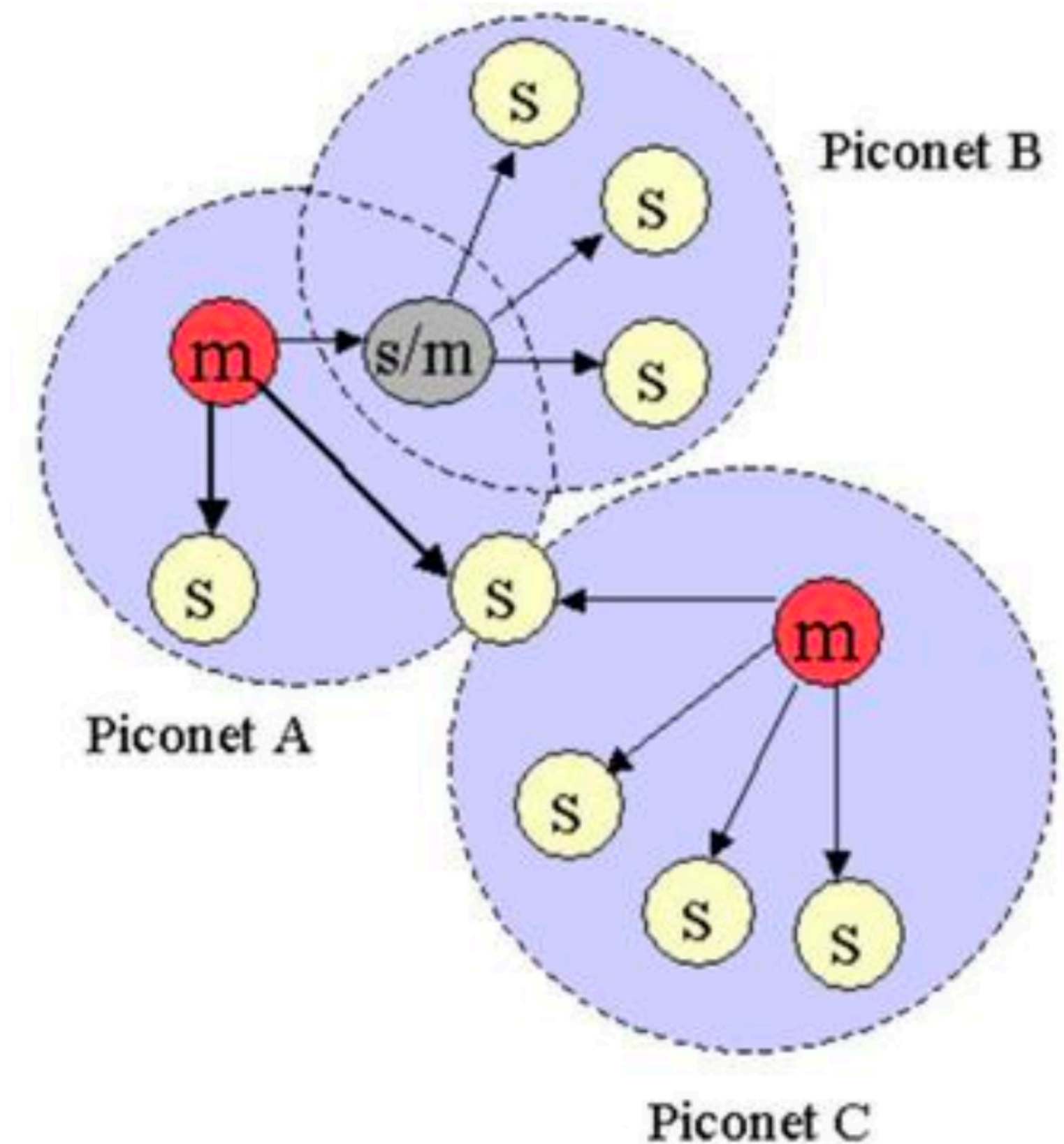
Forming a bluetooth piconet

- The phase in the hopping pattern is determined by the master's clock.
- After adjusting the internal clock according to the master a device may participate in the piconet.
- All active devices are assigned a 3-bit **active member address** (AMA).
- All parked devices use an 8-bit **parked member address** (PMA).
- Devices in stand-by do not need an address.

Bluetooth

Forming a Scatternet

- A device in one piconet can communicate to another device in another piconet, forming a **scatternet**.
- A master in one piconet may be a slave in another piconet.
- Both piconets use a different hopping sequence, always determined by the master of the piconet.
- Bluetooth applies FH-CDMA for separation of piconets.
- A collision occurs if two or more piconets use the same carrier frequency at the same time.

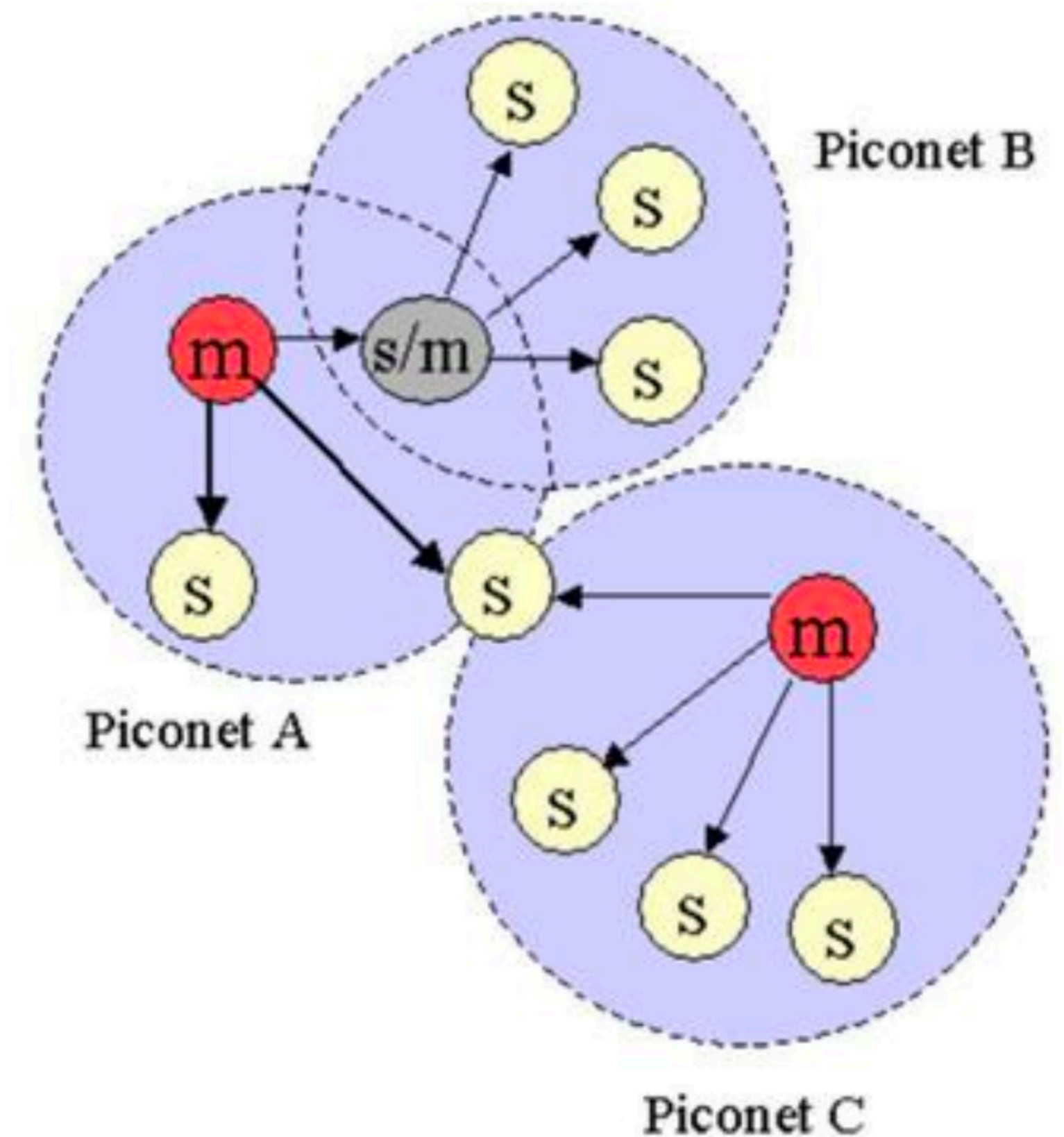


Scatternet

Bluetooth

Forming a Scatternet

- If a device acts as slave in one piconet, it simply starts to synchronize with the hopping sequence of the piconet it wants to join.
- After synchronization, it acts as a slave in this piconet and no longer participates in its former piconet.
- To enable synchronization, a slave has to know the identity of the master that determines the hopping sequence of a piconet.
- Before leaving one piconet, a slave informs the current master that it will be unavailable for a certain amount of time. The remaining devices in the piconet continue to communicate as usual.

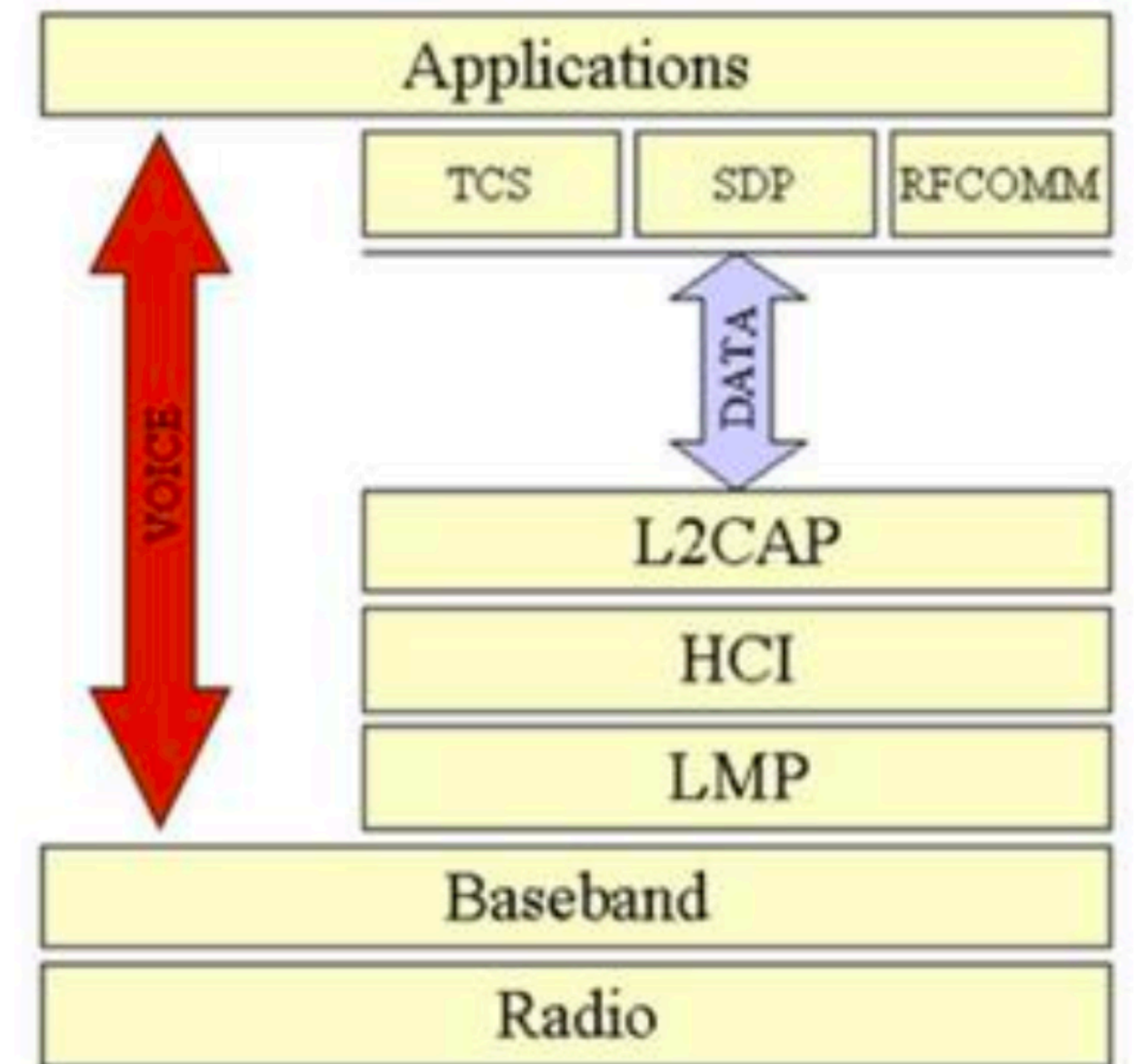


Scatternet

Bluetooth

Protocol Stack

- The Bluetooth protocol stack can be divided into two specifications:
 - **Core specification**, which describes the protocols from physical layer to the data link control together with management functions.
 - **Profile specifications**, which describes many protocols and functions needed to adapt the wireless Bluetooth technology to legacy and new applications.

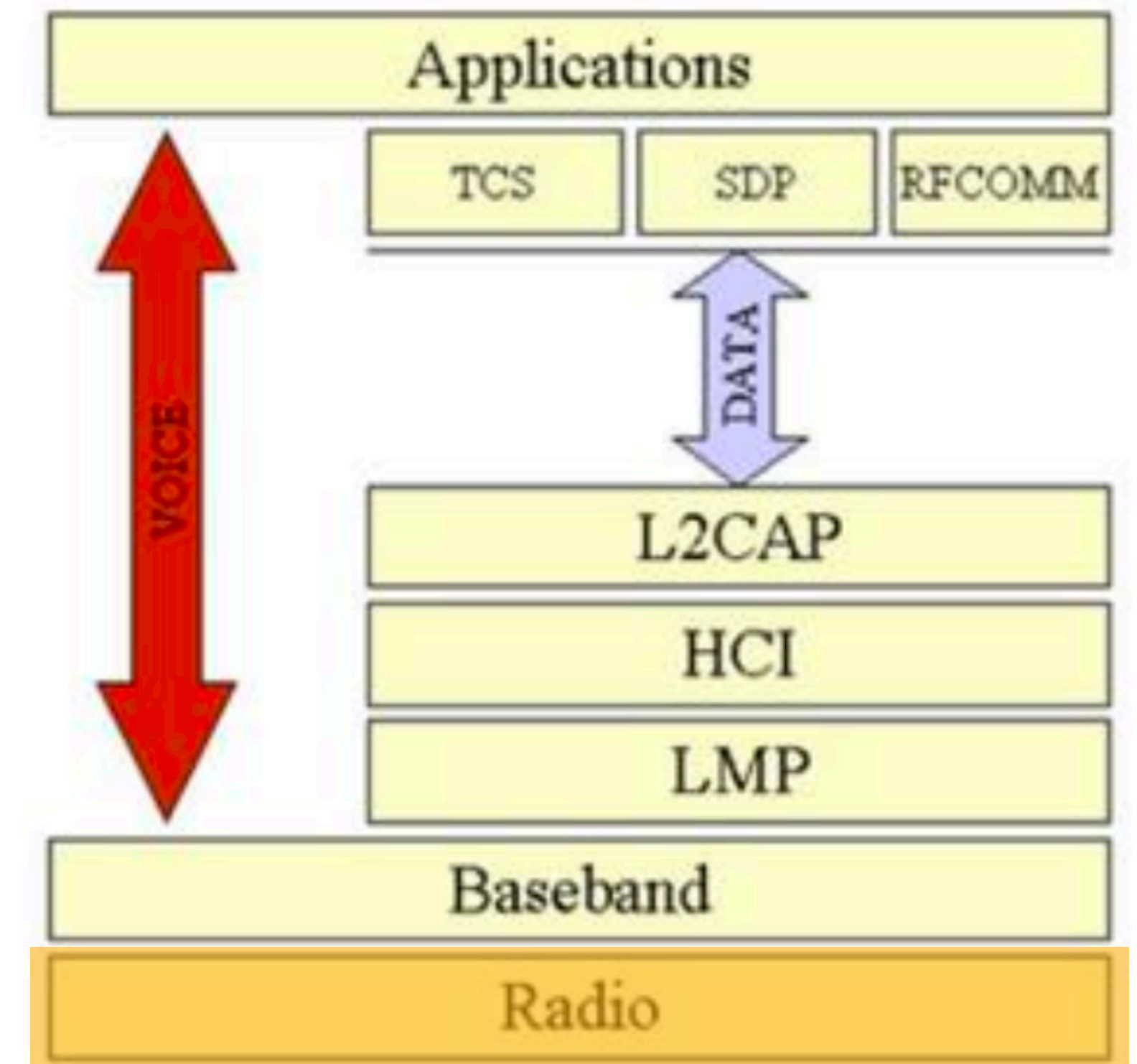


Bluetooth Protocol Stack

Bluetooth

Radio layer

- The radio layer is the physical wireless connection.
- To avoid interference with other devices that communicate in the ISM band, the modulation is based on fast frequency hopping.
- Bluetooth divides the 2.4 GHz frequency band into 79 channels 1 MHz apart (from 2.402 to 2.480 GHz),
- Spread spectrum is used to hop from one channel to another, up to 1600 times a second.
- The standard radio range is 10 cm to 10 m, and can be extended to 100 m by increasing transmission power.

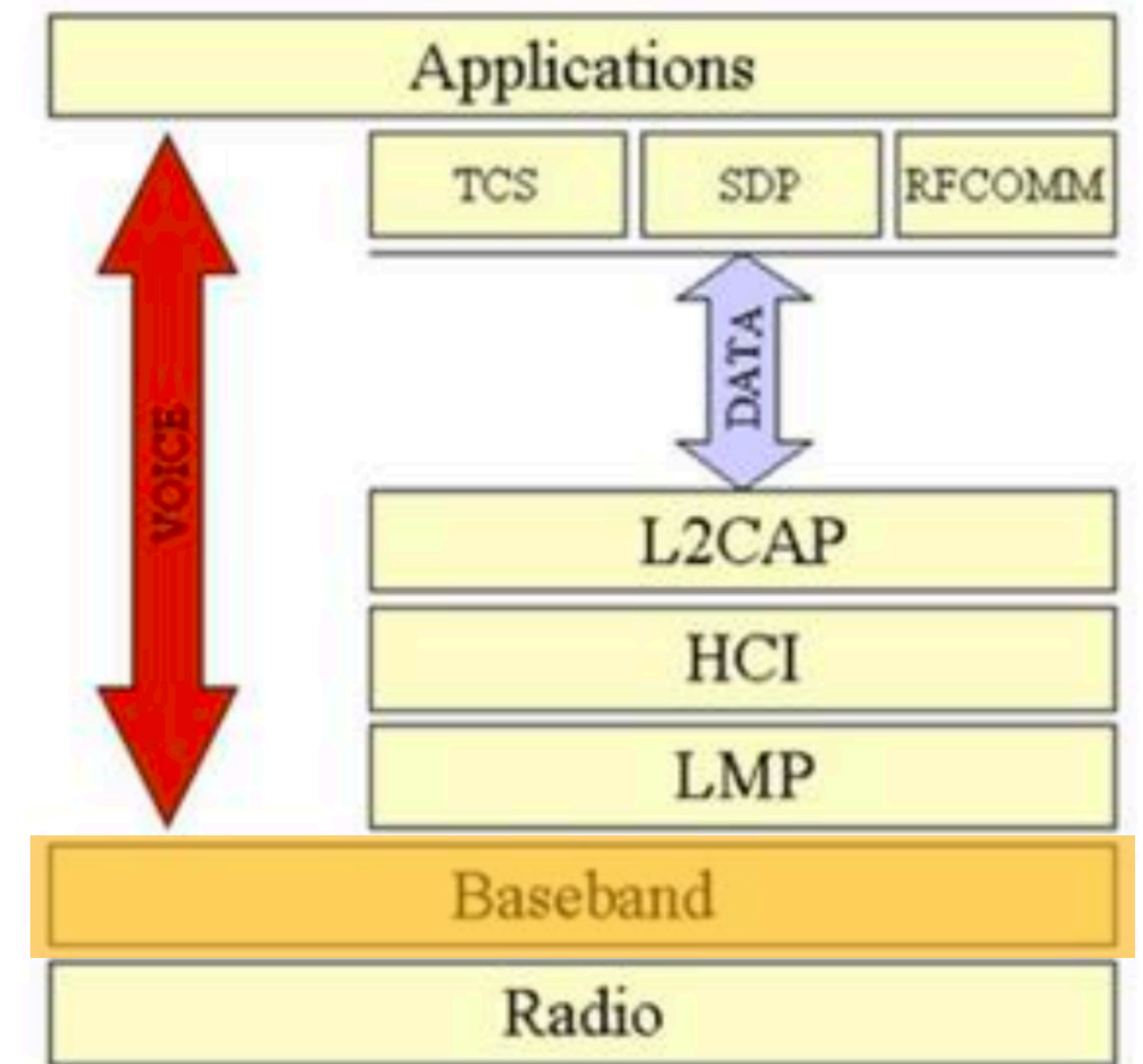


Bluetooth Protocol Stack

Bluetooth

Baseband layer

- The baseband layer is responsible for controlling and sending data packets over the radio link.
- It provides transmission channels for both **data** and **voice**.
- The baseband layer maintains **Synchronous Connection-Oriented** (SCO) links for voice and **Asynchronous Connectionless** (ACL) links for data.
- SCO packets are never retransmitted but ACL packets are, to ensure data integrity.
- SCO links are point-to-point symmetric connections, where time slots are reserved to guarantee timely transmission.
- A slave device is allowed to respond during the time slot immediately following an SCO transmission from the master.

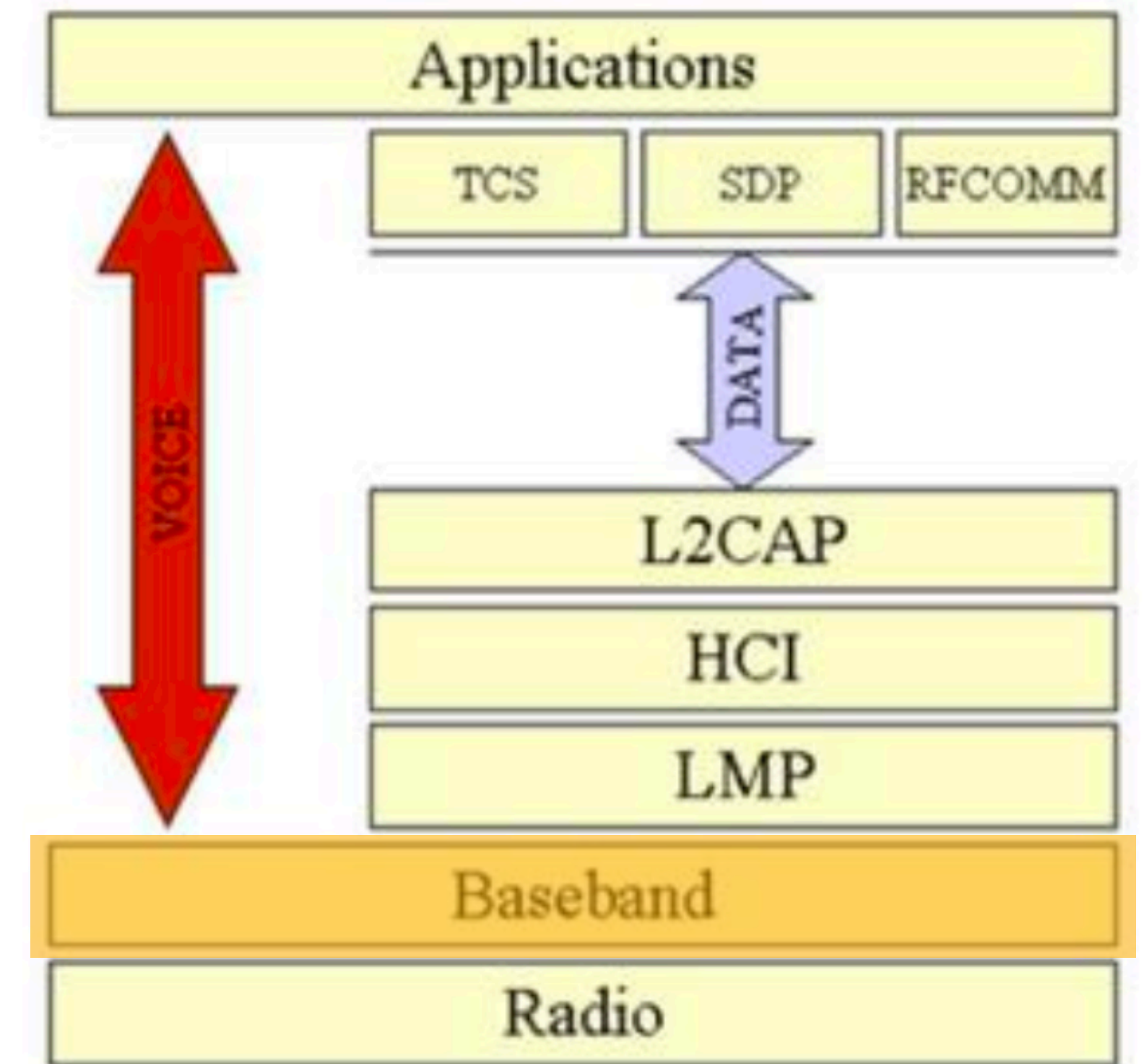


Bluetooth Protocol Stack

Bluetooth

Baseband layer

- Data transmissions on ACL links, on the other hand, are established on a **per-slot basis** (using slots not reserved for SCO links).
- ACL links support point-to- multipoint transmissions.
- After an ACL transmission from the master, only a *slave addressed* specifically may respond during the next time slot; if no device is addressed, the message is treated as a broadcast.

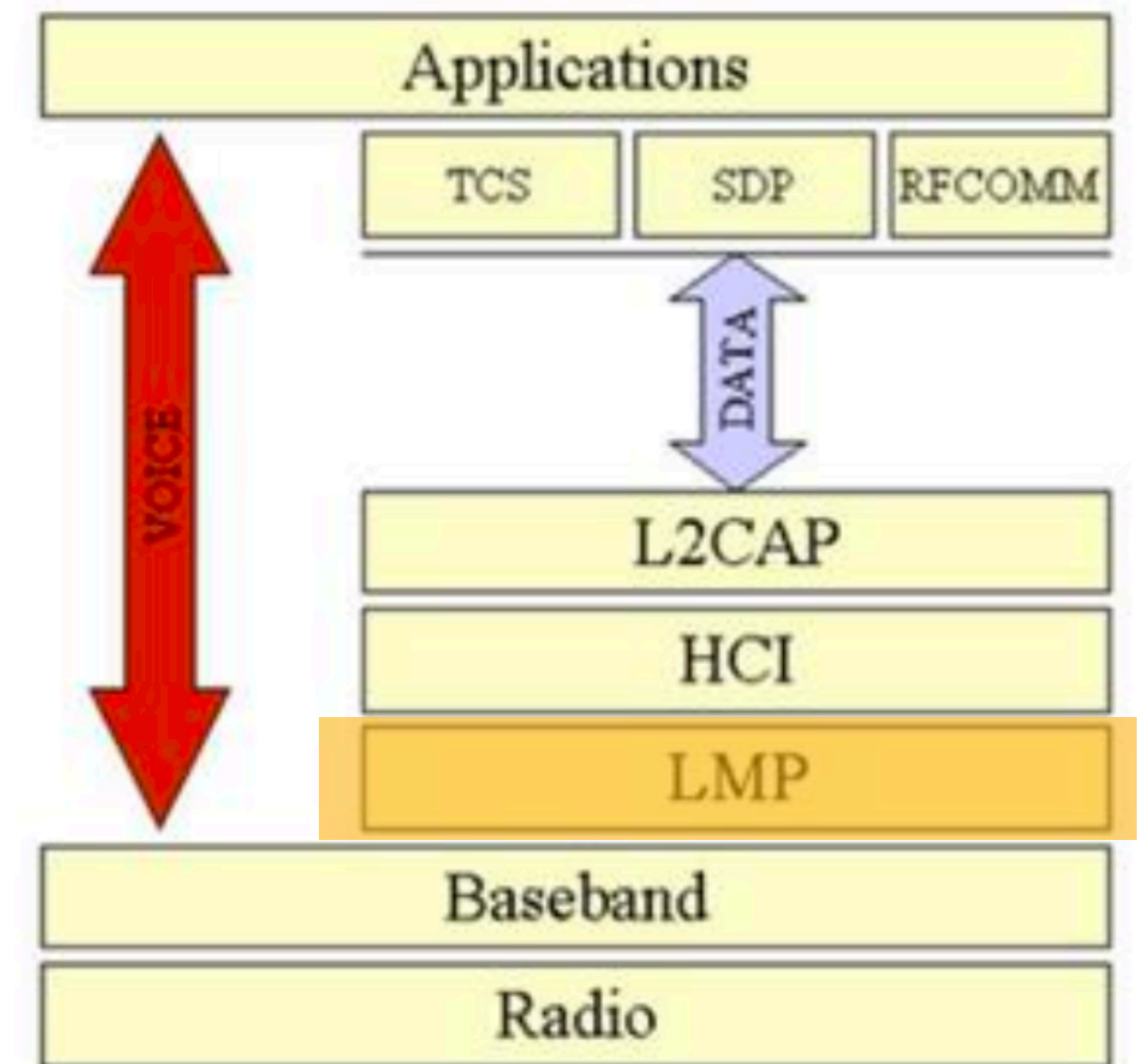


Bluetooth Protocol Stack

Bluetooth

LMP layer

- The **Link Manager Protocol** (LMP) uses the links set up by the baseband to establish connections and manage piconets.
- Responsibilities of the LMP also include:
 - Authentication services
 - Security services
 - Monitoring of service quality

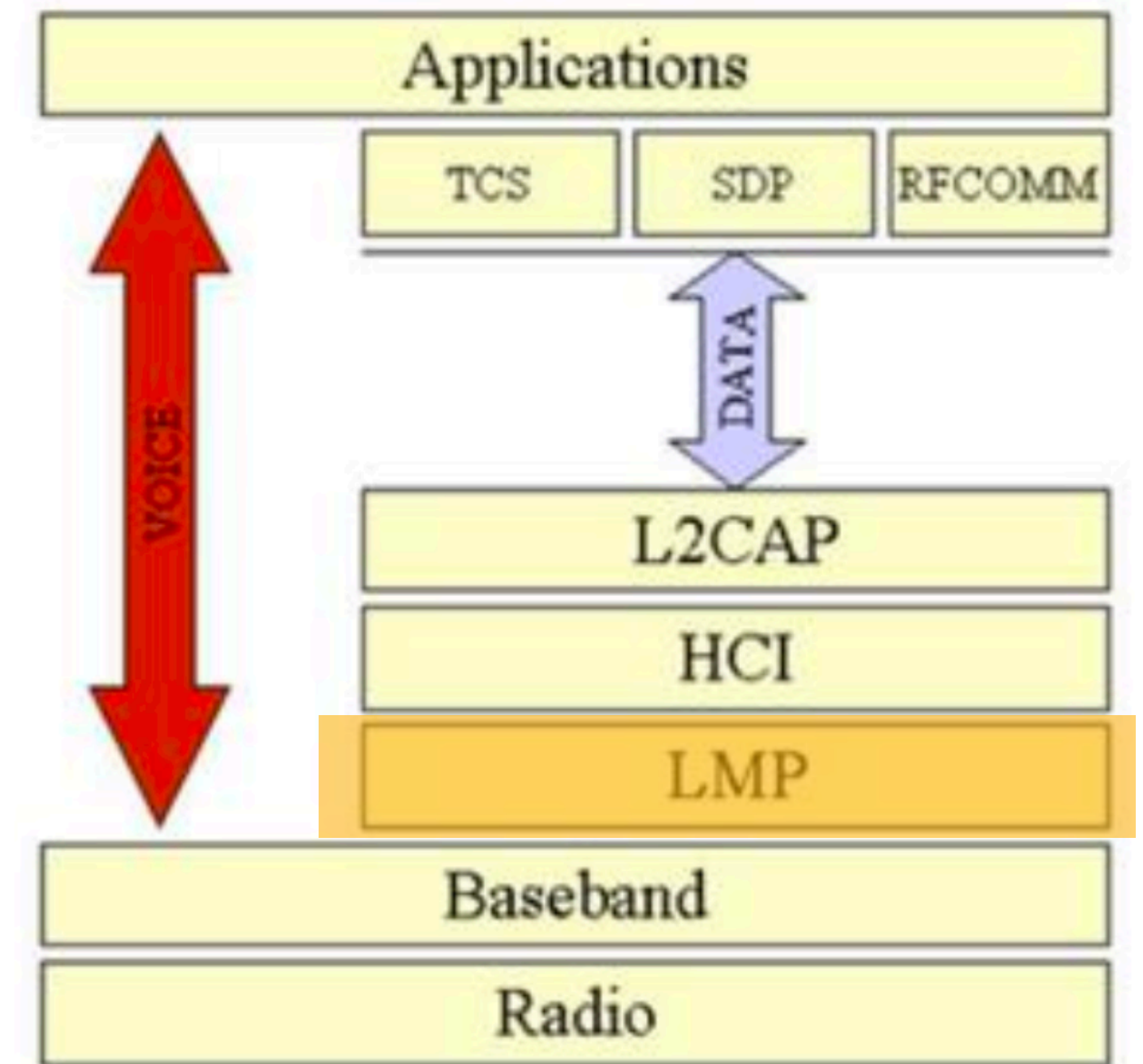


Bluetooth Protocol Stack

Bluetooth

LMP layer

- LMP enhances baseband functionality, but higher layers can still directly access the baseband.
- The following groups of functions are covered by the LMP:
 - **Authentication, pairing, and encryption:**
 - Although basic authentication is handled in the baseband, LMP has to control the exchange of random numbers and signed responses.
 - LMP is not directly involved in the encryption process, but sets the encryption mode (no encryption, point-to-point, or broadcast), key size, and random speed.

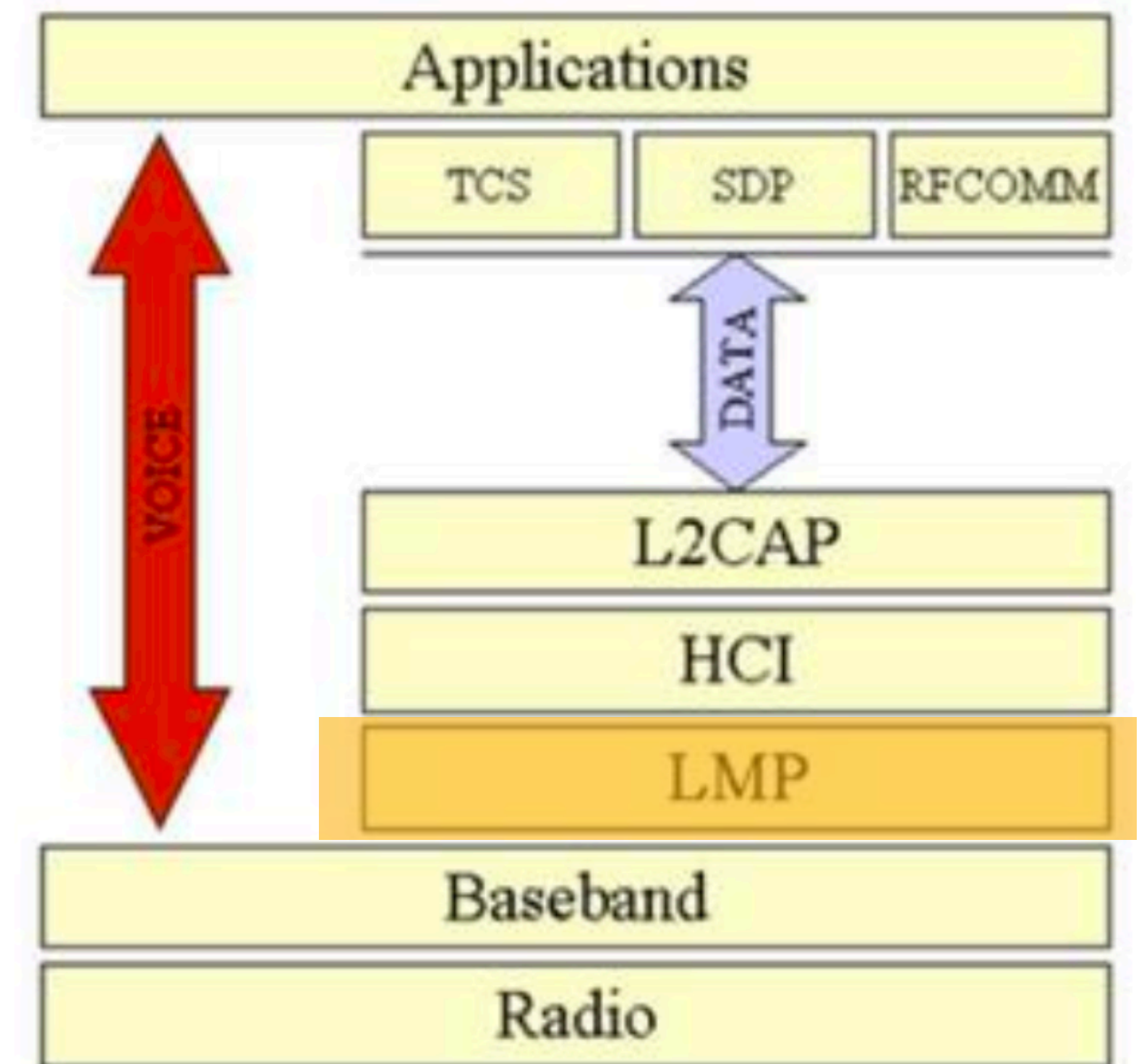


Bluetooth Protocol Stack

Bluetooth

LMP layer

- **Synchronization:**
 - Precise synchronization is of major importance within a Bluetooth network.
 - The clock offset is updated each time a packet is received from the master.
- **Capability negotiation:**
 - Not only the version of the LMP can be exchanged but also information about the supported features.
 - Not all Bluetooth devices will support all features that are described in the standard, so devices have to agree the usage of, e.g., multi-slot packets, encryption, SCO links, voice encoding, park/sniff/hold mode, HV2/HV3 packets etc.

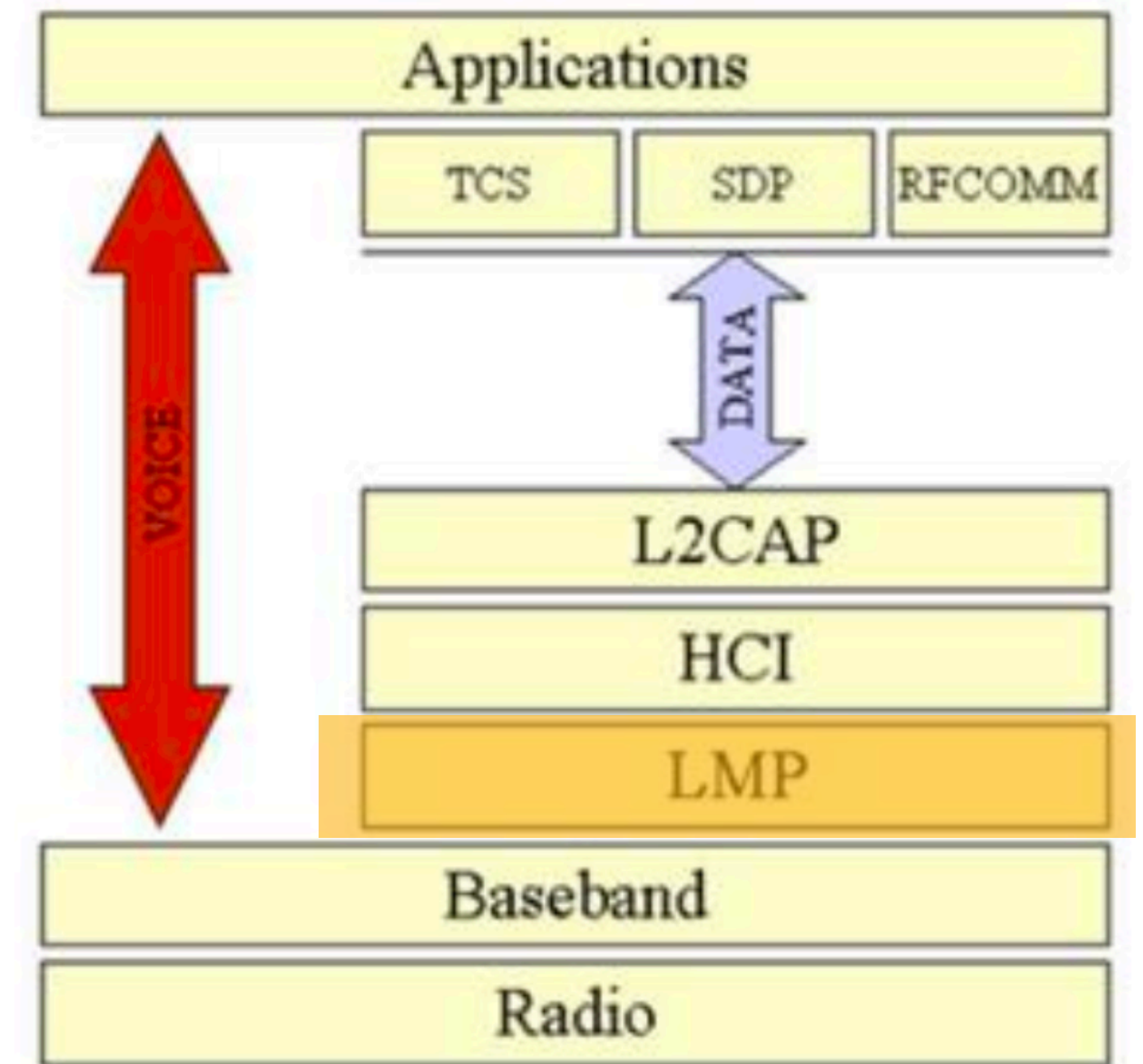


Bluetooth Protocol Stack

Bluetooth

LMP layer

- **Quality of service negotiation:**
 - Different parameters control the QoS of a Bluetooth device at lower layers.
 - The poll interval, i.e., the maximum time between transmissions from a master to a particular slave, controls the latency and transfer capacity.
 - A master can also limit the number of slots available for slaves' answers to increase its own bandwidth.

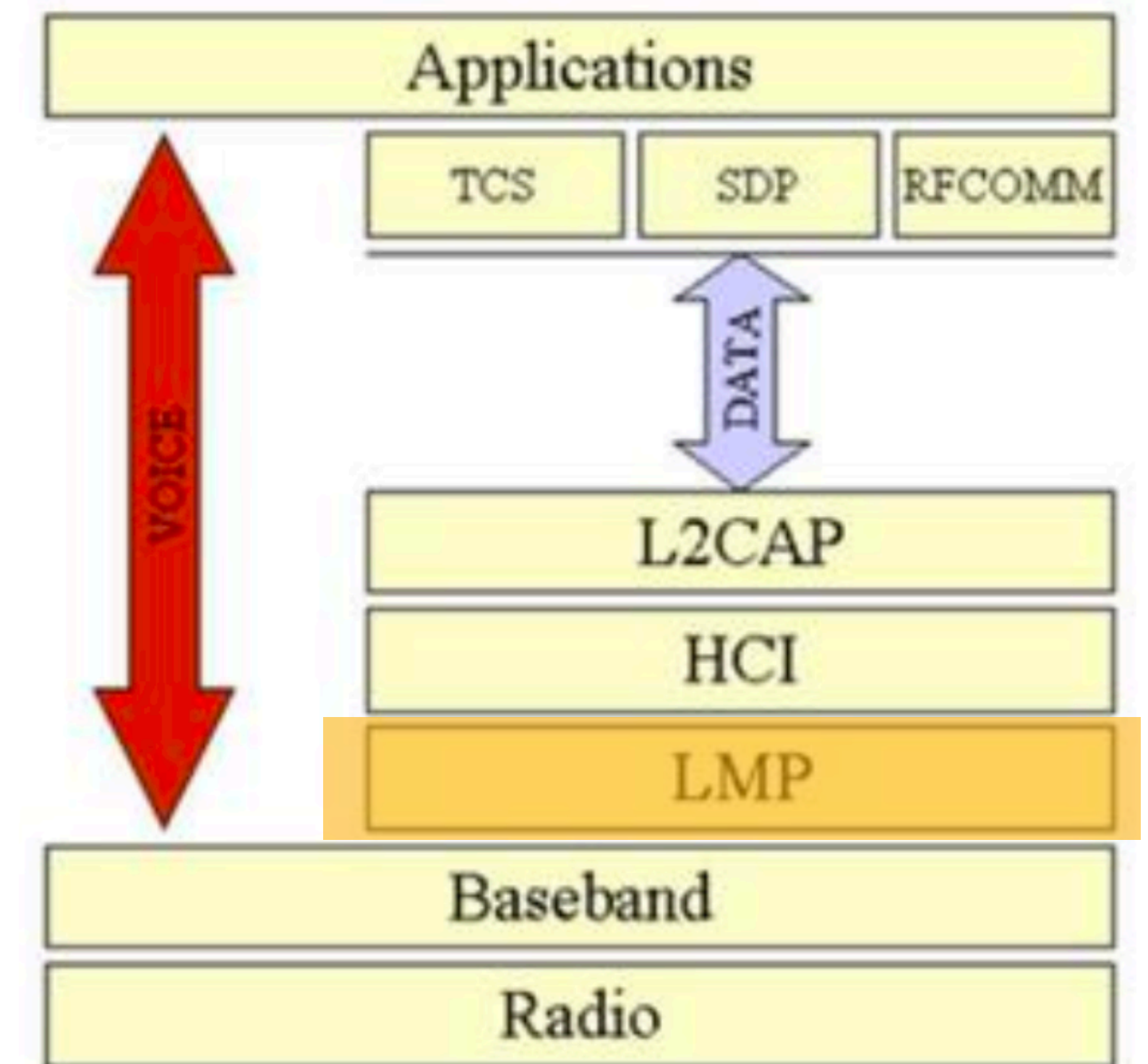


Bluetooth Protocol Stack

Bluetooth

LMP layer

- **Power control:**
 - A Bluetooth device can measure the received signal strength.
 - Depending on this signal level the device can direct the sender of the measured signal to increase or decrease its transmit power.
- **Link supervision:**
 - LMP has to control the activity of a link, it may set up new SCO links, or it may declare the failure of a link.
- **State and transmission mode change:**
 - Devices might switch the master/slave role, detach themselves from a connection, or change the operating mode.

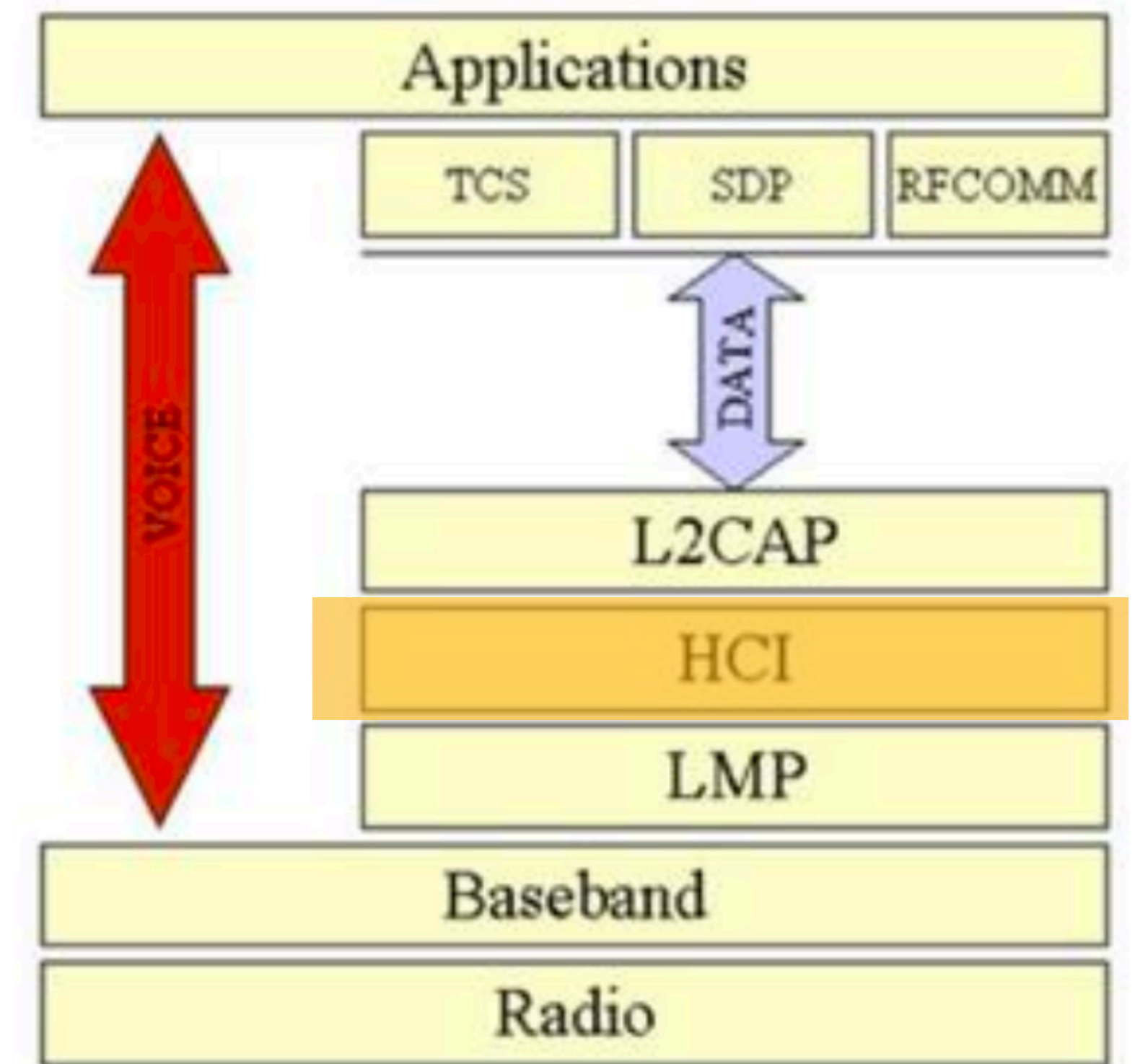


Bluetooth Protocol Stack

Bluetooth

HCI layer

- The **Host Controller Interface (HCI)** is the dividing line between software and hardware.
- The L2CAP and layers above it are mostly implemented in software and the LMP and lower layers are mostly implemented in hardware.
- The HCI may not be required.
- The L2CAP may be accessed directly by the application, or through certain support protocols provided to ease the burden on application programmers.

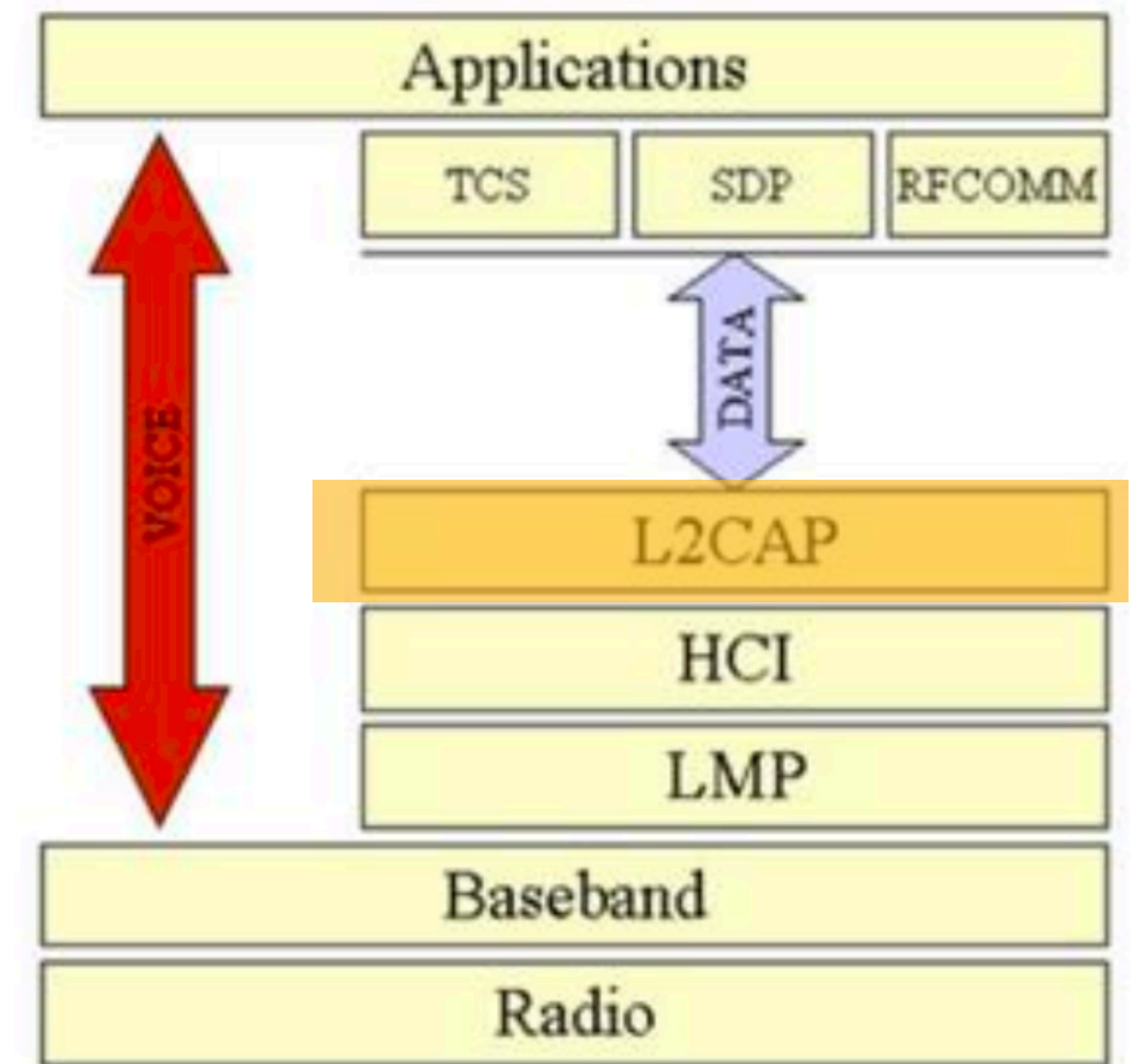


Bluetooth Protocol Stack

Bluetooth

L2CAP layer

- The **Logical Link Control and Adaptation Protocol** (L2CAP) receives application data and adapts it to the Bluetooth format.
- L2CAP is a data link control protocol on top of the baseband layer offering logical channels between Bluetooth devices with QoS properties.
- Quality of Service (QoS) parameters are exchanged at this layer.
- L2CAP is available for ACLs only.

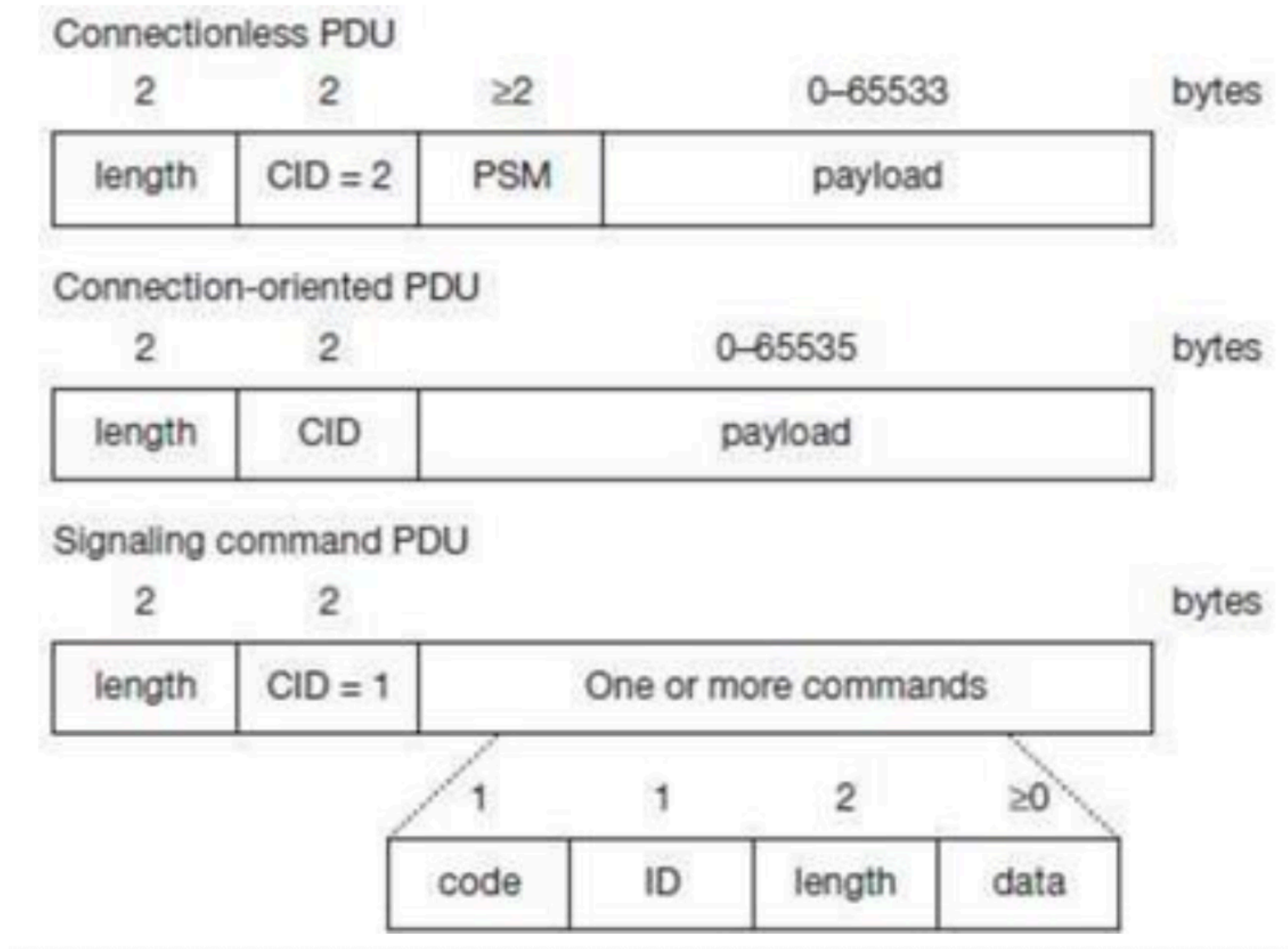


Bluetooth Protocol Stack

Bluetooth

L2CAP layer

- L2CAP provides three different types of logical channels that are transported via the ACL between master and slave:
 - **Connectionless:** These unidirectional channels are typically used for broadcasts from a master to its slave(s).
 - **Connection-oriented:** Each channel of this type is bi-directional and supports QoS flow specifications for each direction. These flow specs follow RFC 1363 and define average/peak data rate, maximum burst size, latency, and jitter.
 - **Signaling:** A logical channel which is used for exchanging signaling messages between L2CAP entities.

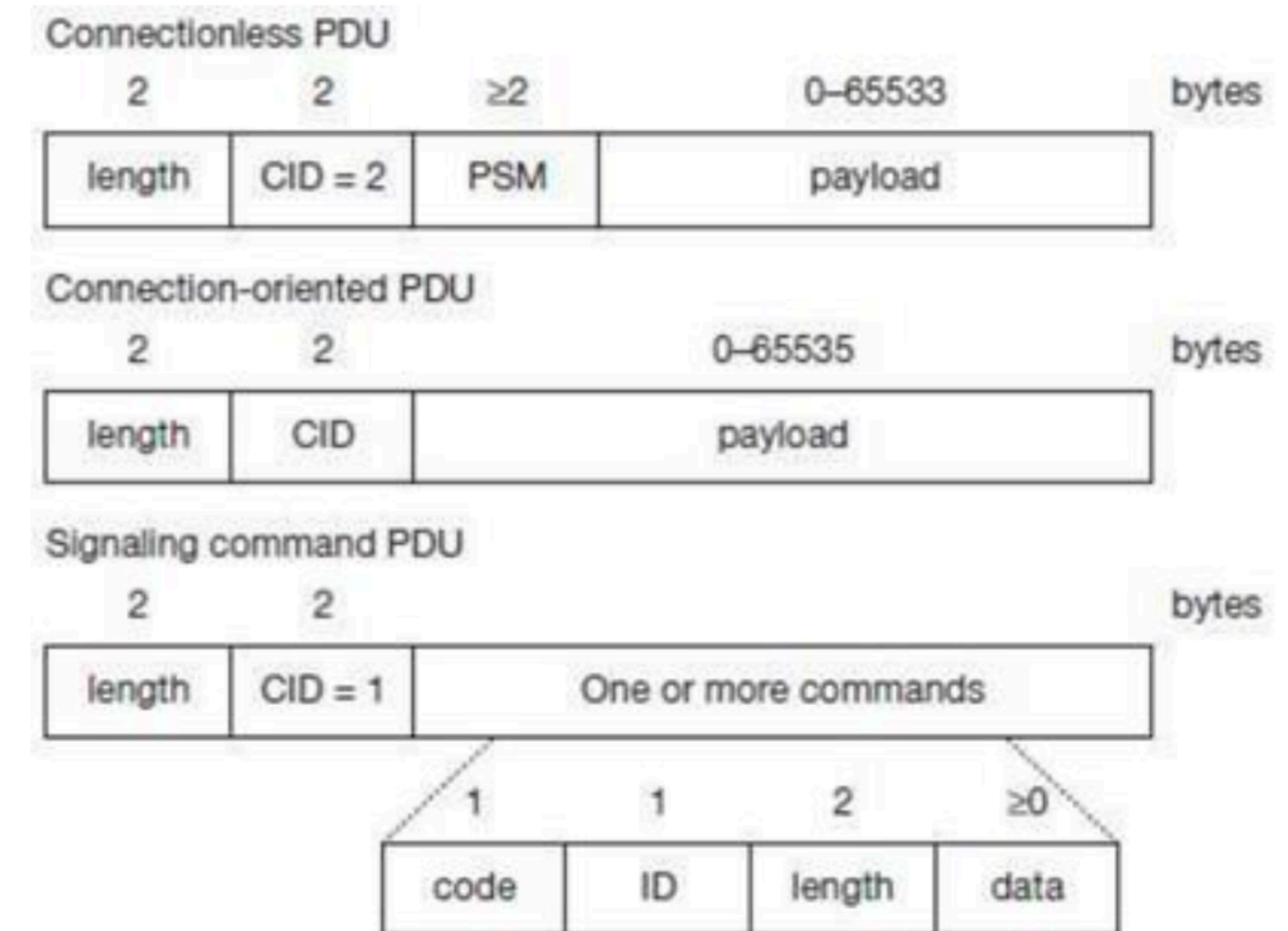


The three packet types

Bluetooth

L2CAP layer

- Each channel can be identified by its channel identifier (CID).
- Signaling channels always use a CID value of 1, a CID value of 2 is reserved for connectionless channels.
- For connection-oriented channels a unique CID (≥ 64) is dynamically assigned at each end of the channel to identify the connection.
- There are three packet types belonging to the three logical channel types.

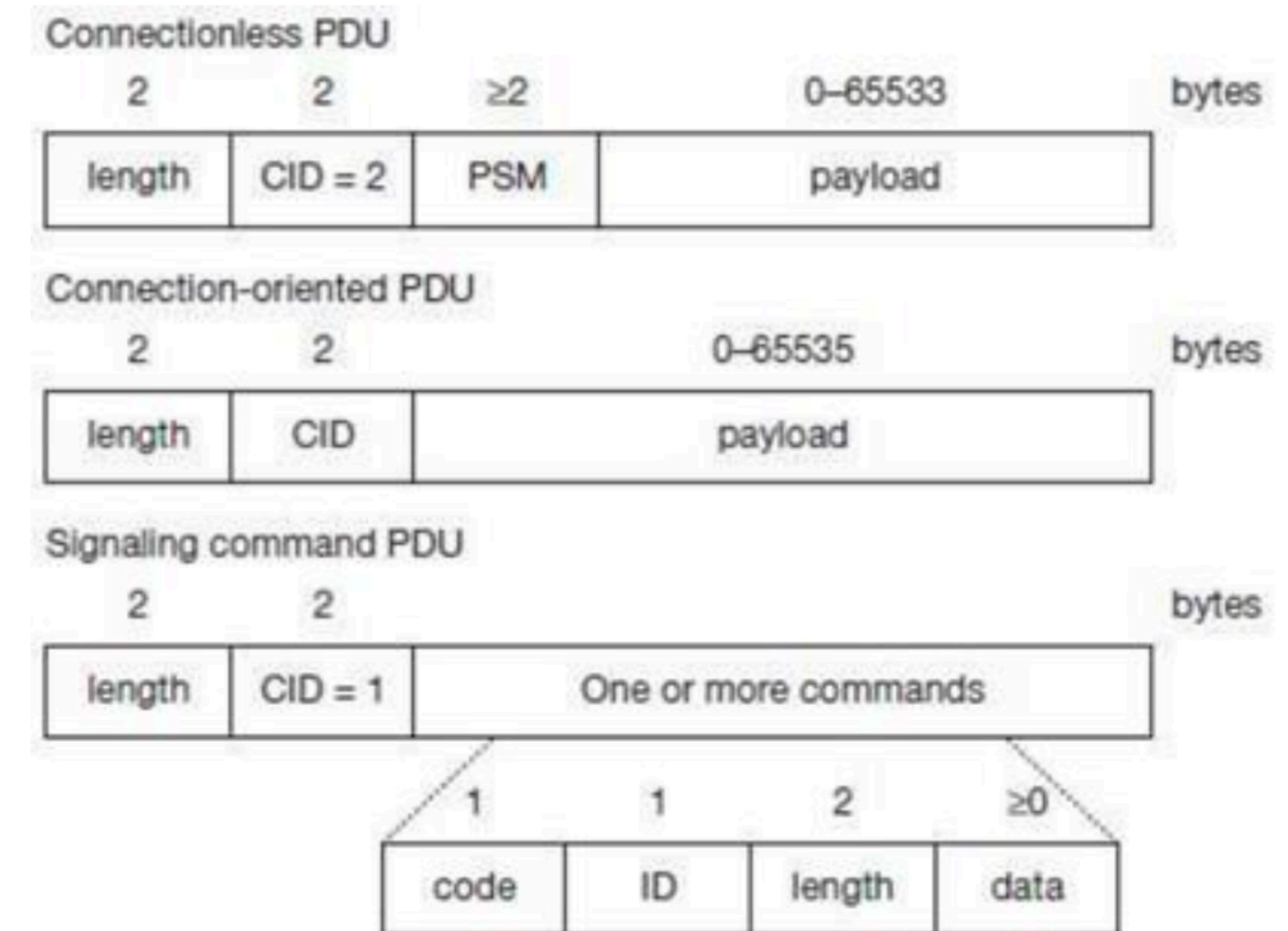


The three packet types

Bluetooth

L2CAP layer

- The length field indicates the length of the payload (plus PSM for connectionless PDUs).
- The CID has the multiplexing/demultiplexing function.
 - For connectionless PDUs a **protocol/service multiplexor (PSM)** field is needed to identify the higher layer recipient for the payload.
 - For connection-oriented PDUs the CID already fulfills this function.
 - Several PSM values have been defined, e.g., 1 (SDP), 3 (RFCOMM), 5 (TCS-BIN). Values above 4096 can be assigned dynamically.
- The payload of the signaling PDU contains one or more commands.
- Each command has its own code (e.g., for command reject, connection request, disconnection response etc.) and an ID that matches a request with its reply.
- The length field indicates the length of the data field for this command.

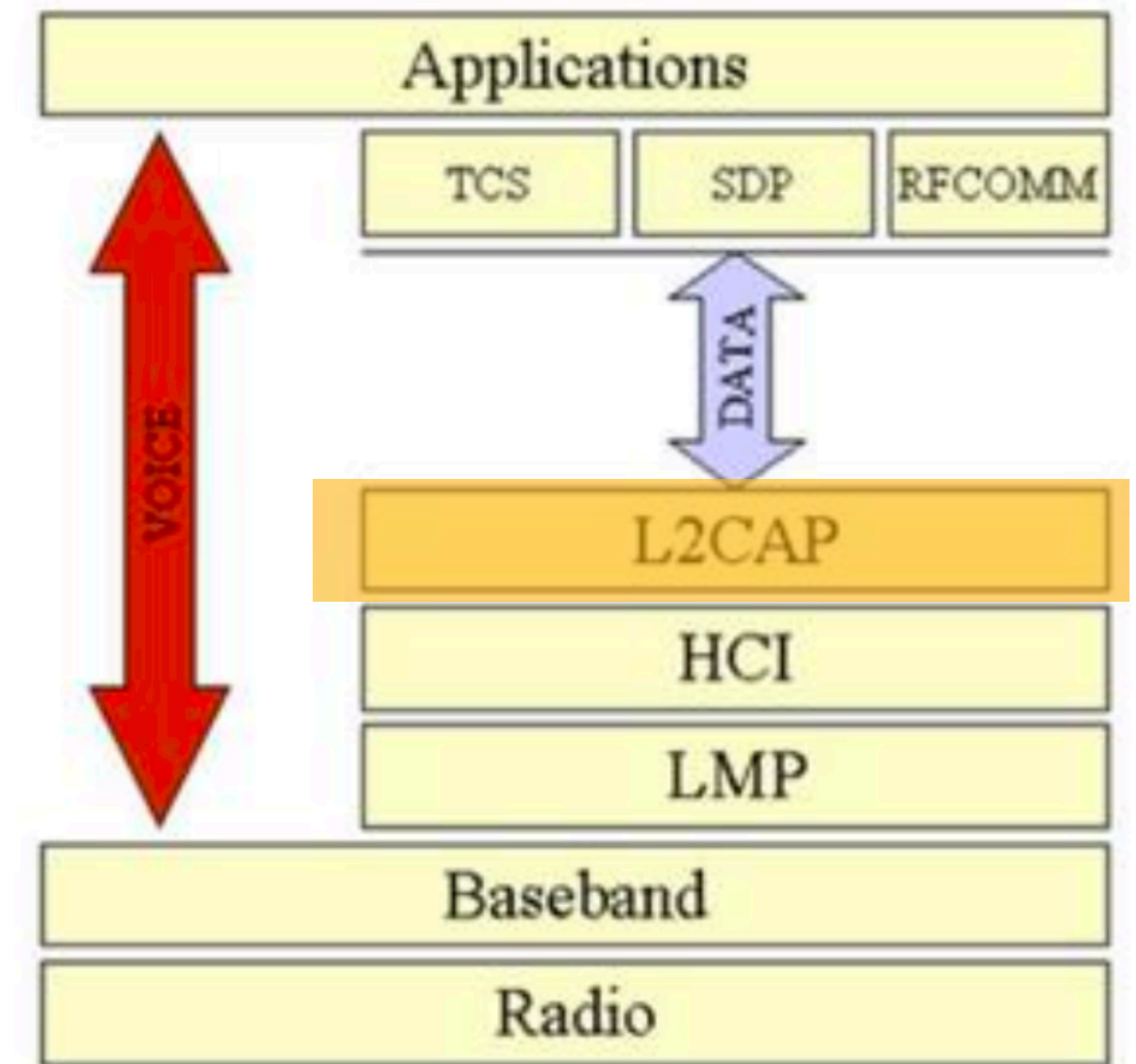


The three packet types

Bluetooth

L2CAP layer

- The L2CAP layer provides:
 - Protocol multiplexing
 - Flow specification
 - Group management
 - Segmentation
 - Reassembly
- Depending on the baseband capabilities, large packets have to be chopped into smaller segments.

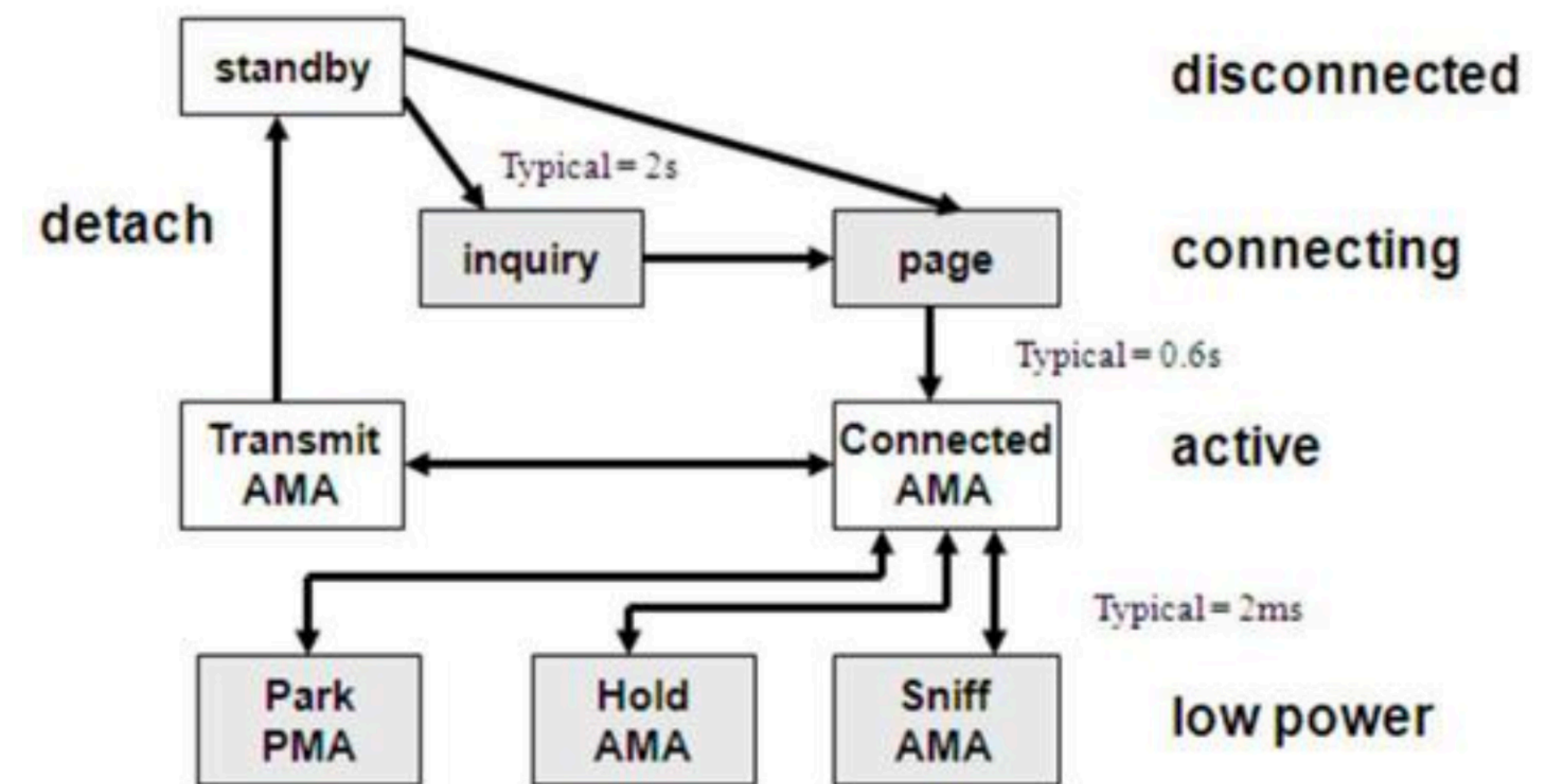


Bluetooth Protocol Stack

Bluetooth

Low-power states

- Bluetooth defines several low-power states for a device.
- Every device, which is currently not participating in a piconet (and not switched off), is in standby mode.
- Standby mode is a low-power mode where only the native clock is running.
- The next step towards the inquiry mode can happen in two different ways.
- A user of the device wants to scan for other devices in the radio range.
- The device starts the inquiry procedure by sending an inquiry access code (IAC) that is common to all Bluetooth devices. The IAC is broadcast over 32 so-called wake-up carriers in turn.
- Devices in standby may enter the inquiry mode periodically to search for IAC messages on the wake-up carriers. As soon as a device detects an inquiry it returns a packet containing its device address and timing information required by the master to initiate a connection.
- From that moment on, the device acts as slave.

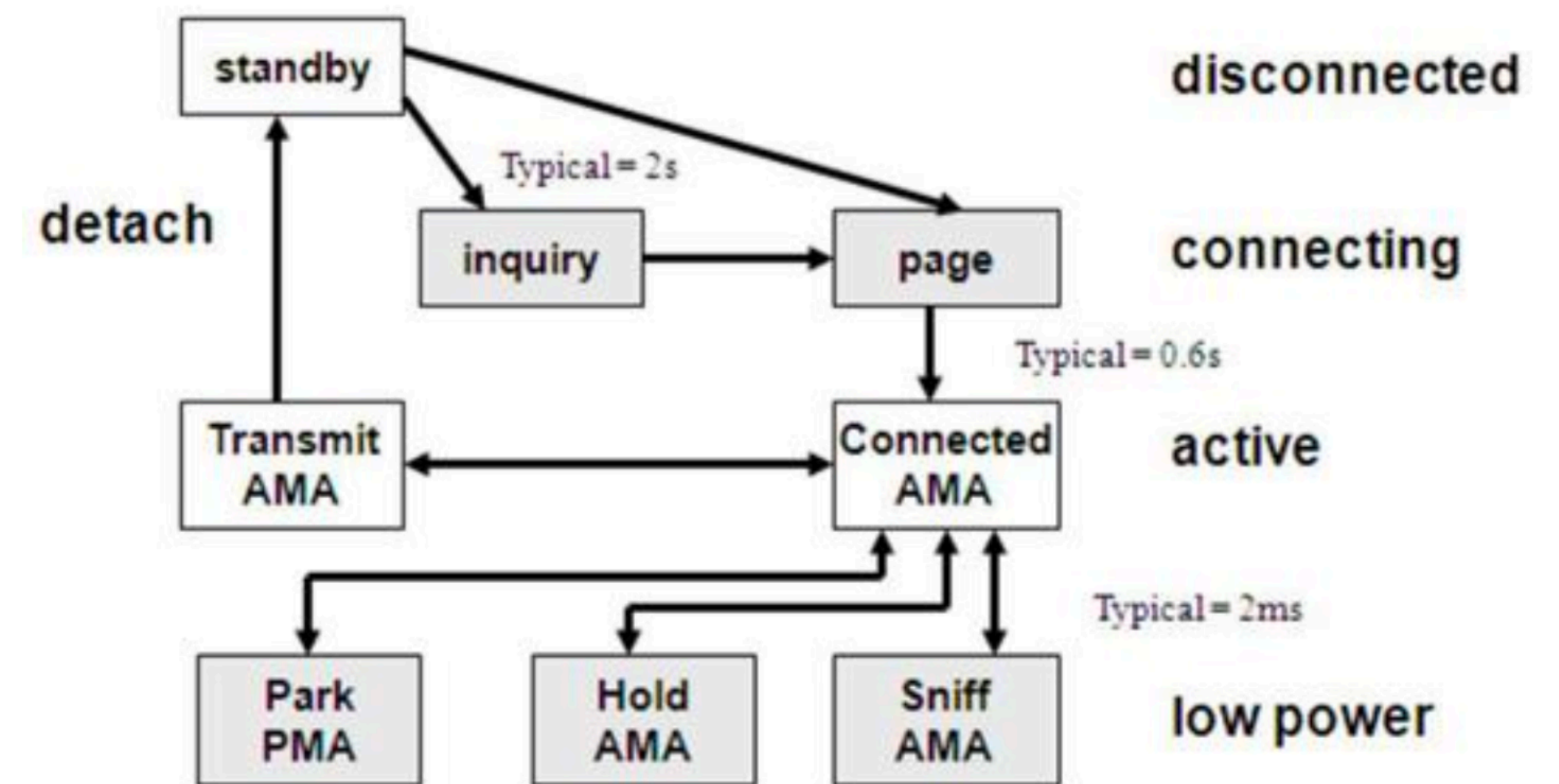


Major baseband states of a Bluetooth device

Bluetooth

Low-power states

- If inquiry was successful, a device enters the page mode.
- The inquiry phase is not coordinated, so it may take a while before the inquiry is successful.
- After a while, a Bluetooth device sees all the devices in its radio range.

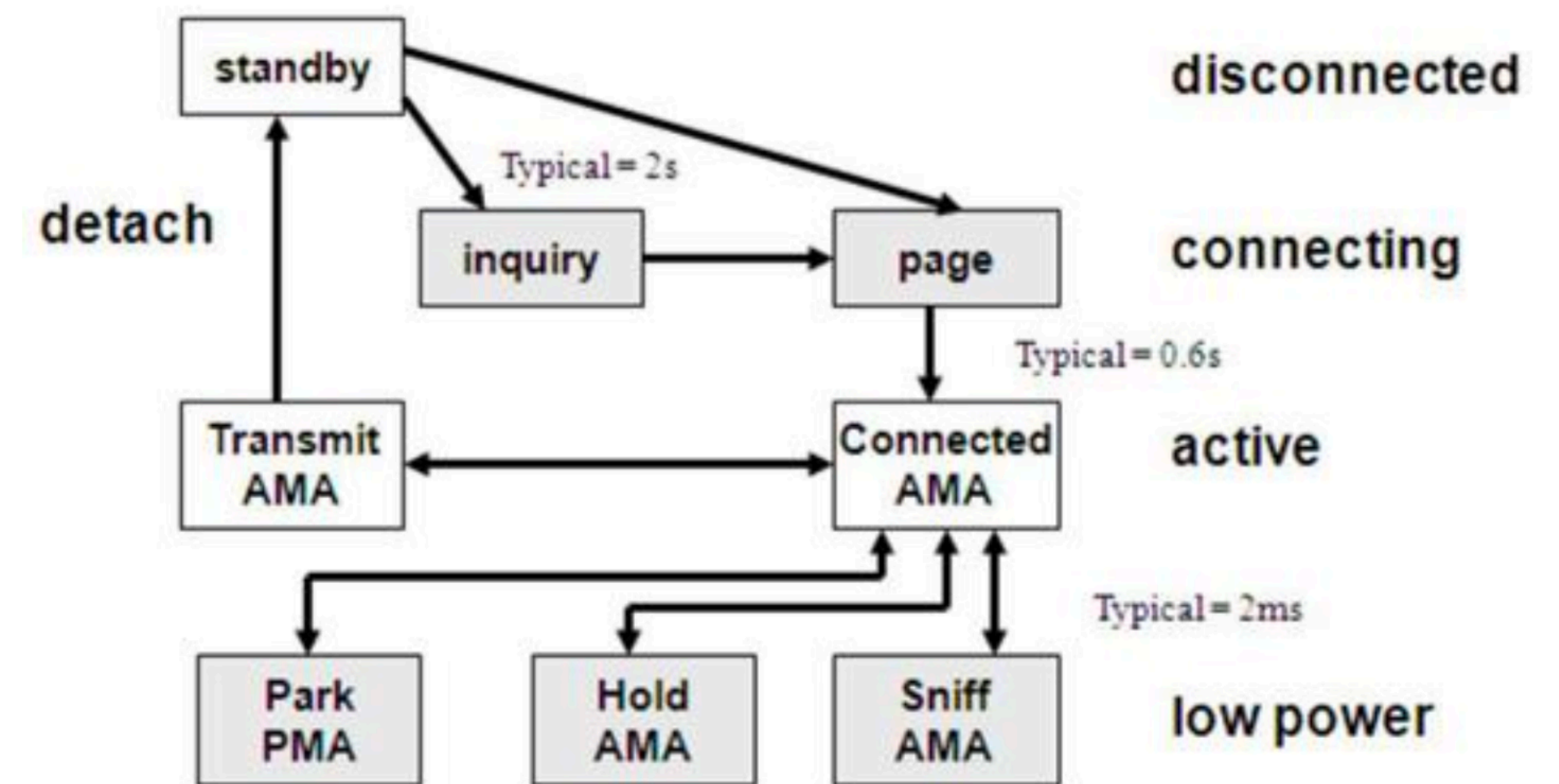


Major baseband states of a Bluetooth device

Bluetooth

Low-power states

- During the page state two different roles are defined.
- After finding all required devices the master is able to set up connections to each device, i.e., setting up a piconet.
- As soon as a device synchronizes to the hopping pattern of the piconet it also enters the connection state.
- The connection state comprises the active state and the low power states: park, sniff, and hold.
- In the active state the slave participates in the piconet by listening, transmitting, and receiving. ACL and SCO links can be used.
- A master periodically synchronizes with all slaves.
- All devices being active must have the 3-bit active member address (AMA).

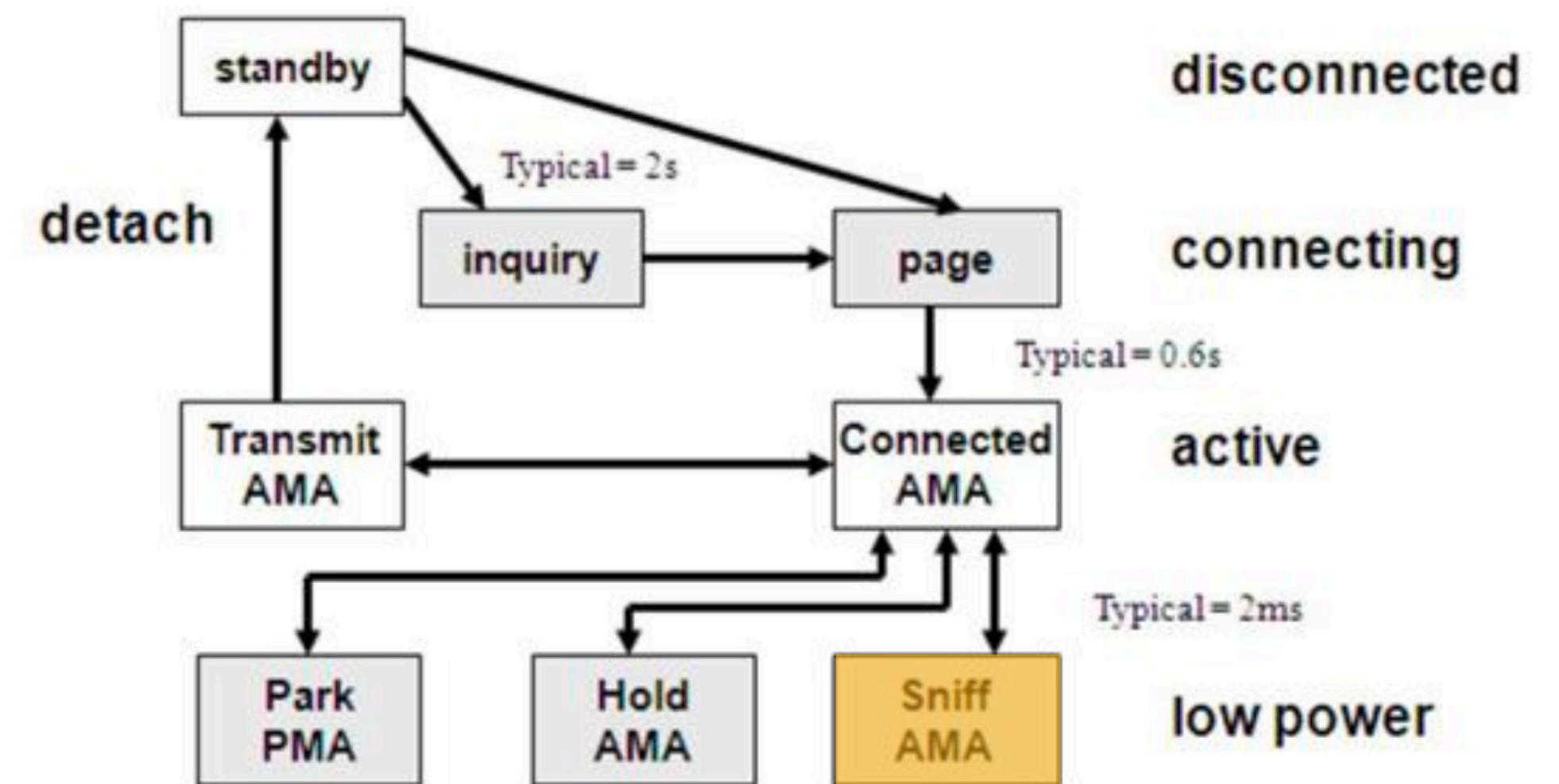


Major baseband states of a Bluetooth device

Bluetooth

Low-power states

- To save battery power, a Bluetooth device can go into one of three low power states:
- **Sniff state:** The sniff state has the highest power consumption of the low power states.
 - Here, the device listens to the piconet at a reduced rate (not on every other slot as is the case in the active state).
 - The interval for listening into the medium can be programmed and is application dependent.
 - The master designates a reduced number of slots for transmission to slaves in sniff state. However, the device keeps its AMA.

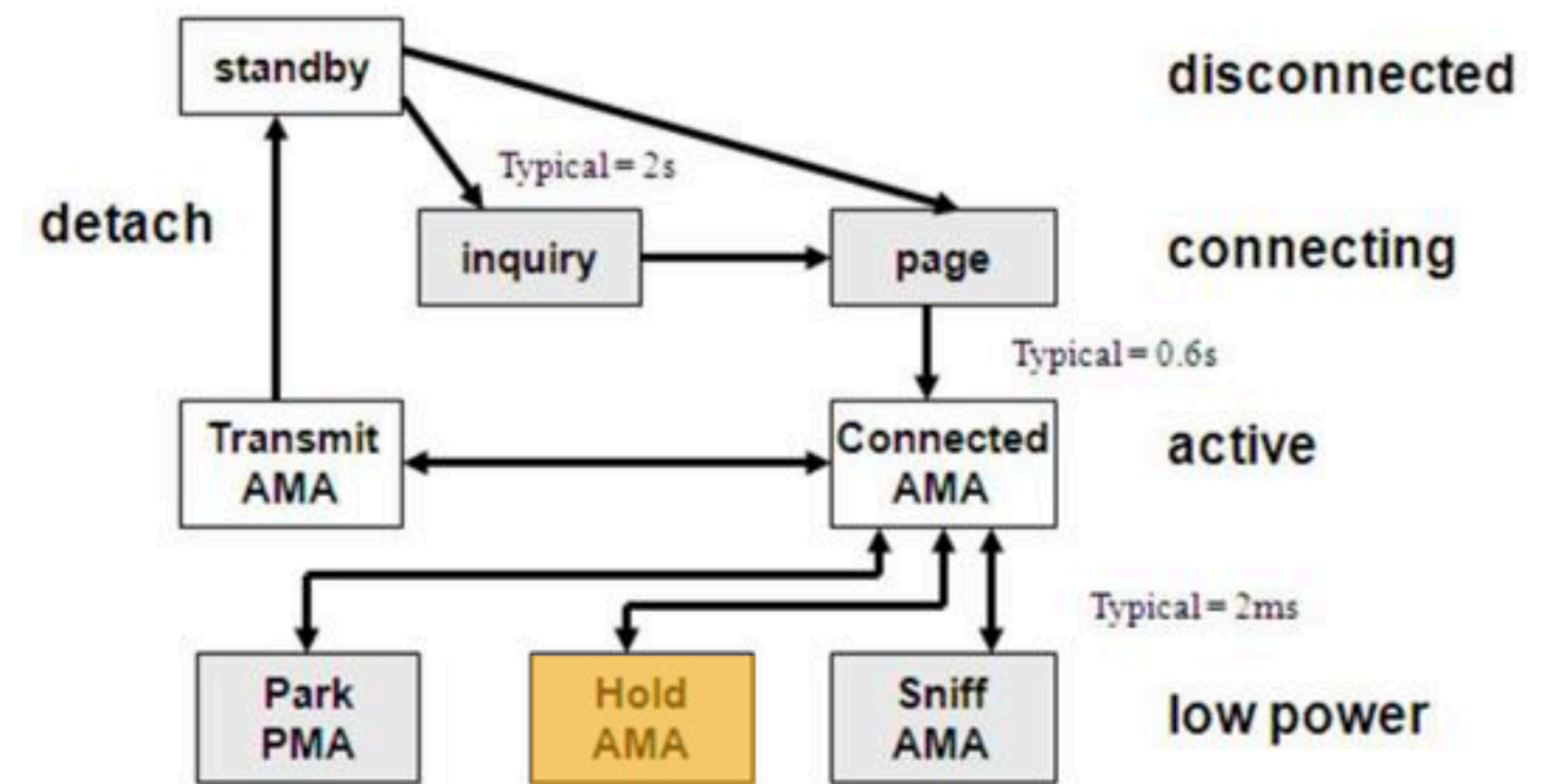


Major baseband states of a Bluetooth device

Bluetooth

Low-power states

- **Hold state:** The device does not release its AMA but stops ACL transmission.
- A slave may still exchange SCO packets.
- If there is no activity in the piconet, the slave may either reduce power consumption or participate in another piconet.

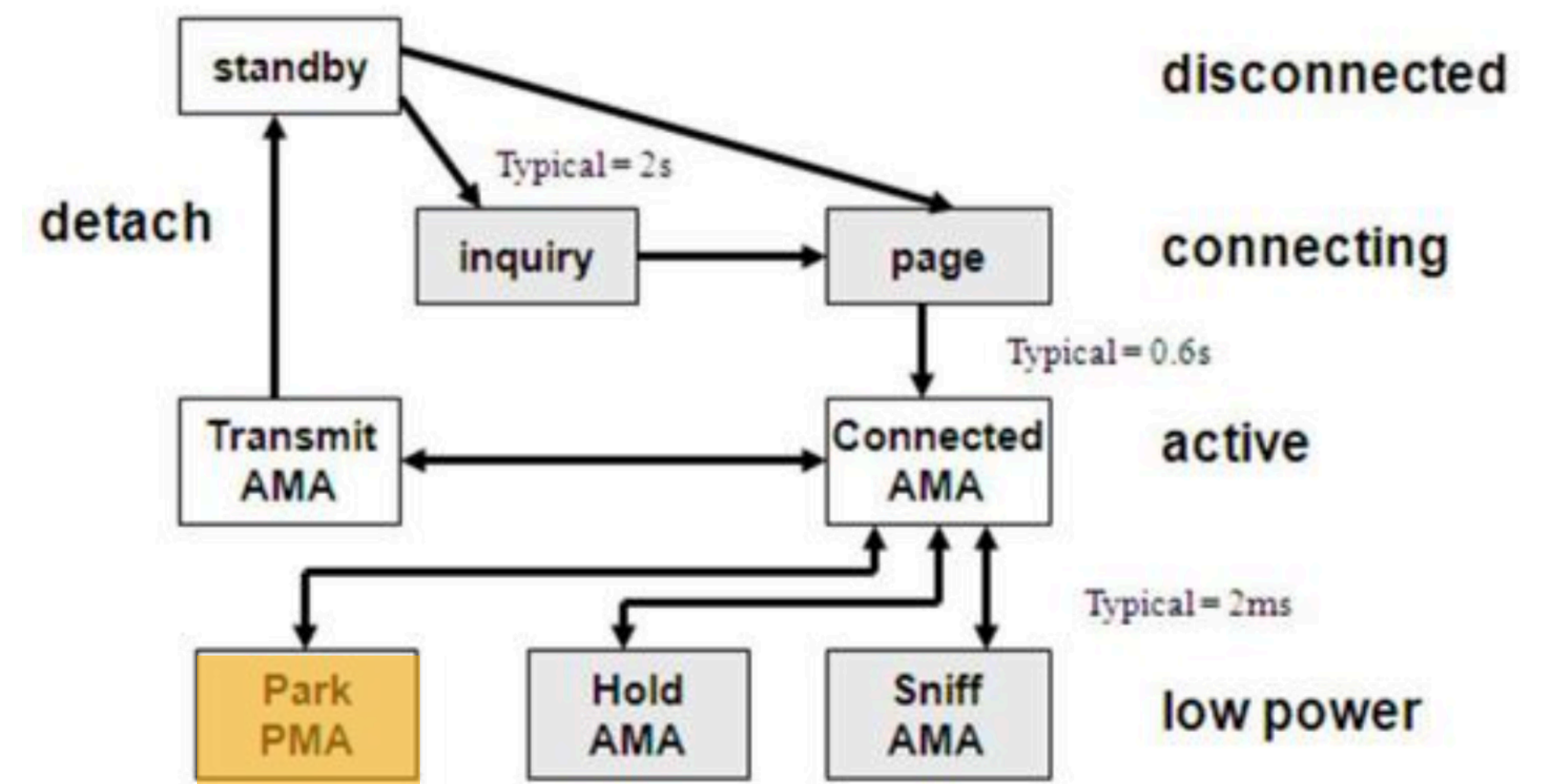


Major baseband states of a Bluetooth device

Bluetooth

Low-power states

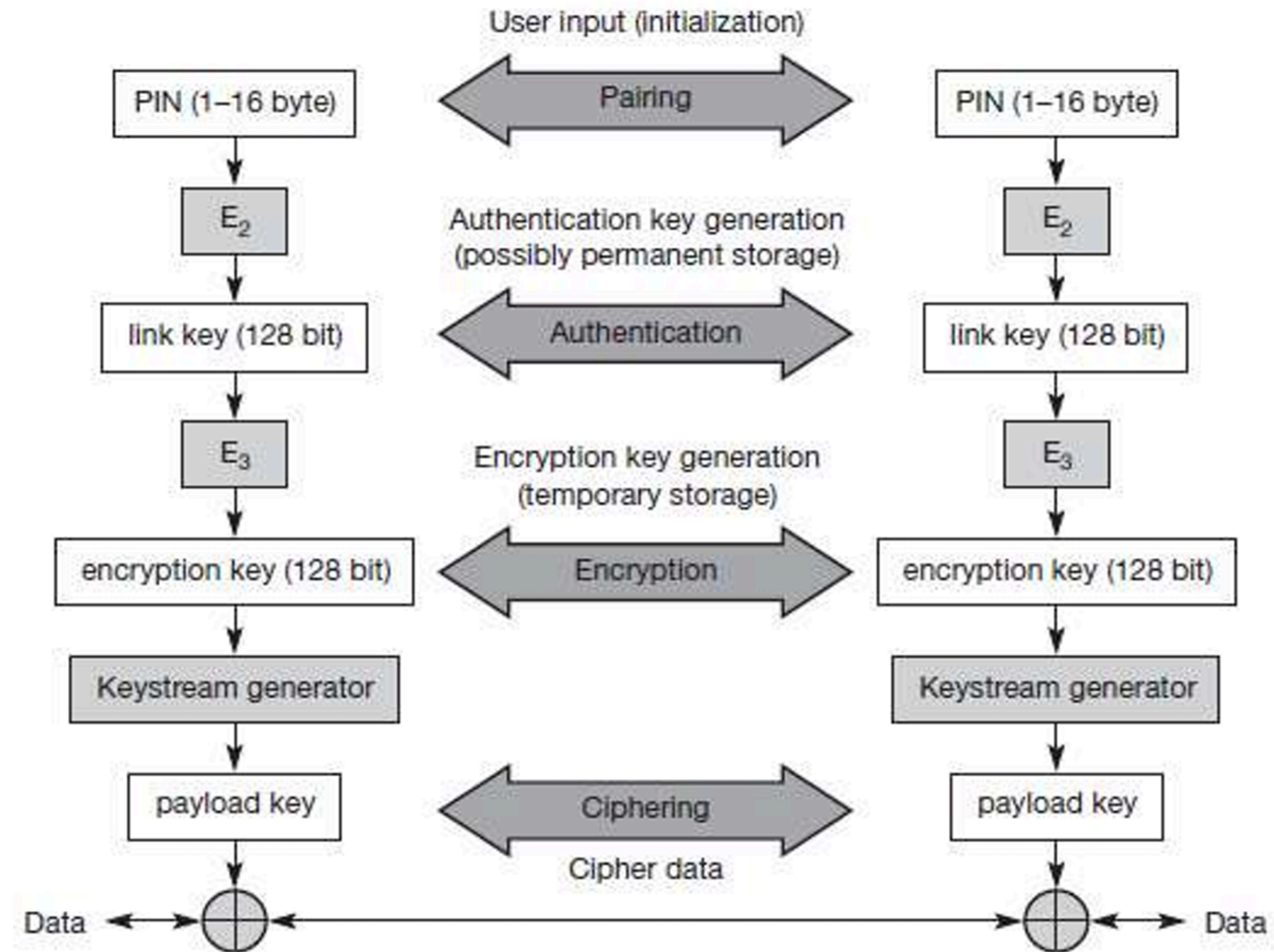
- **Park state:** In this state the device has the lowest duty cycle and the lowest power consumption.
- The device releases its AMA and receives a parked member address (PMA).
- The device is still a member of the piconet, but gives room for another device to become active (AMA is only 3 bit, PMA 8 bit).
- Parked devices are still FH synchronized and wake up at certain beacon intervals for re-synchronization.
- All PDUs sent to parked slaves are broadcast.



Major baseband states of a Bluetooth device

Bluetooth Security

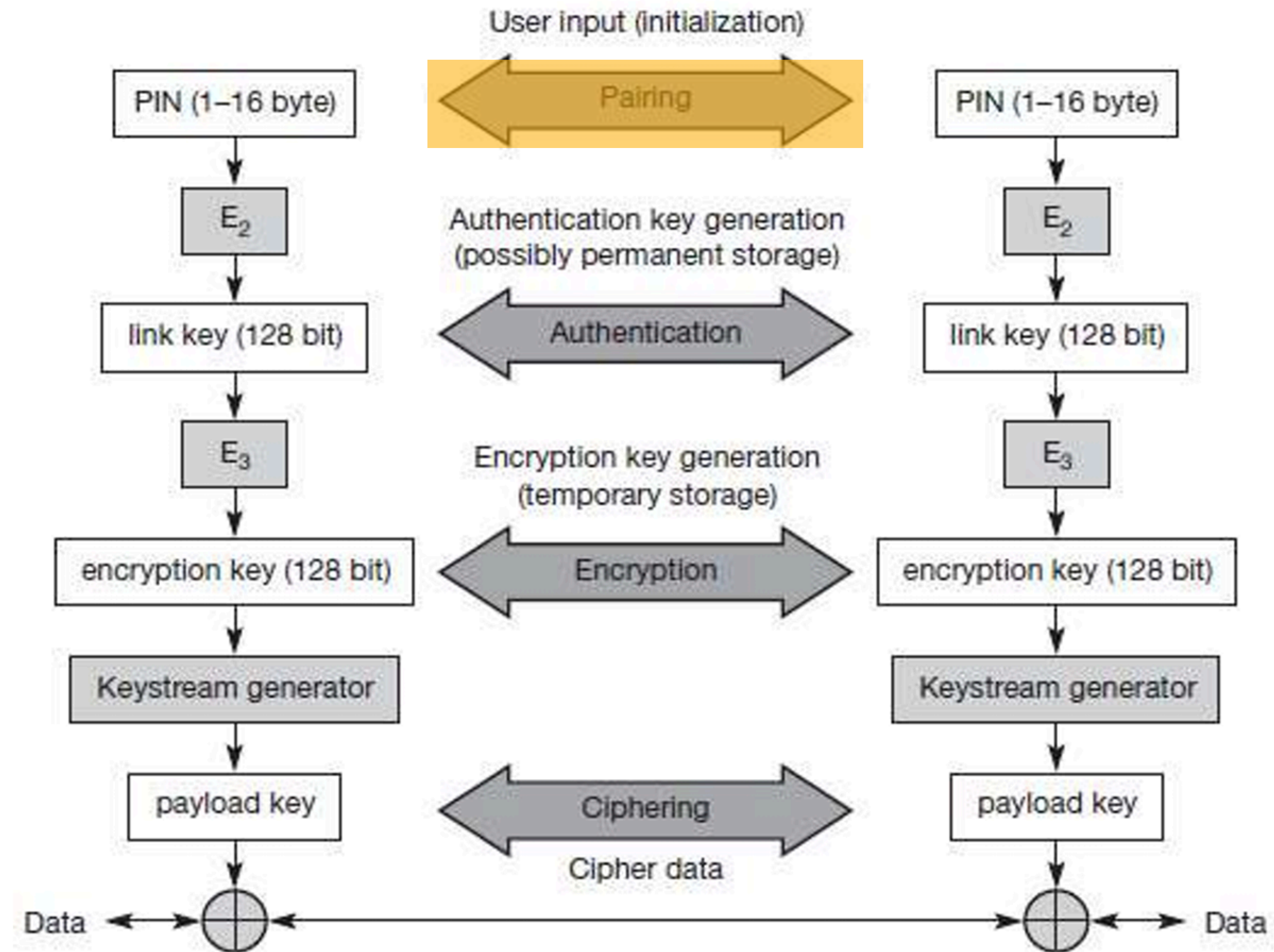
- Bluetooth offers security infrastructure starting from authentication, key exchange to encryption.
- Besides, frequency-hopping scheme with 1600 hops/sec is employed.
- All of this makes the system difficult to eavesdrop.
- The main security features offered by Bluetooth include a **challenge response routine** for authentication, a **stream cipher** for encryption, and a **session key generation**.
- Each connection may require a one-way, two-way, or no authentication using the challenge-response routine.
- The security algorithms use the public identity of a device, a secret private user key, and an internally generated random key as input parameters.



Bluetooth security architecture

Bluetooth Security

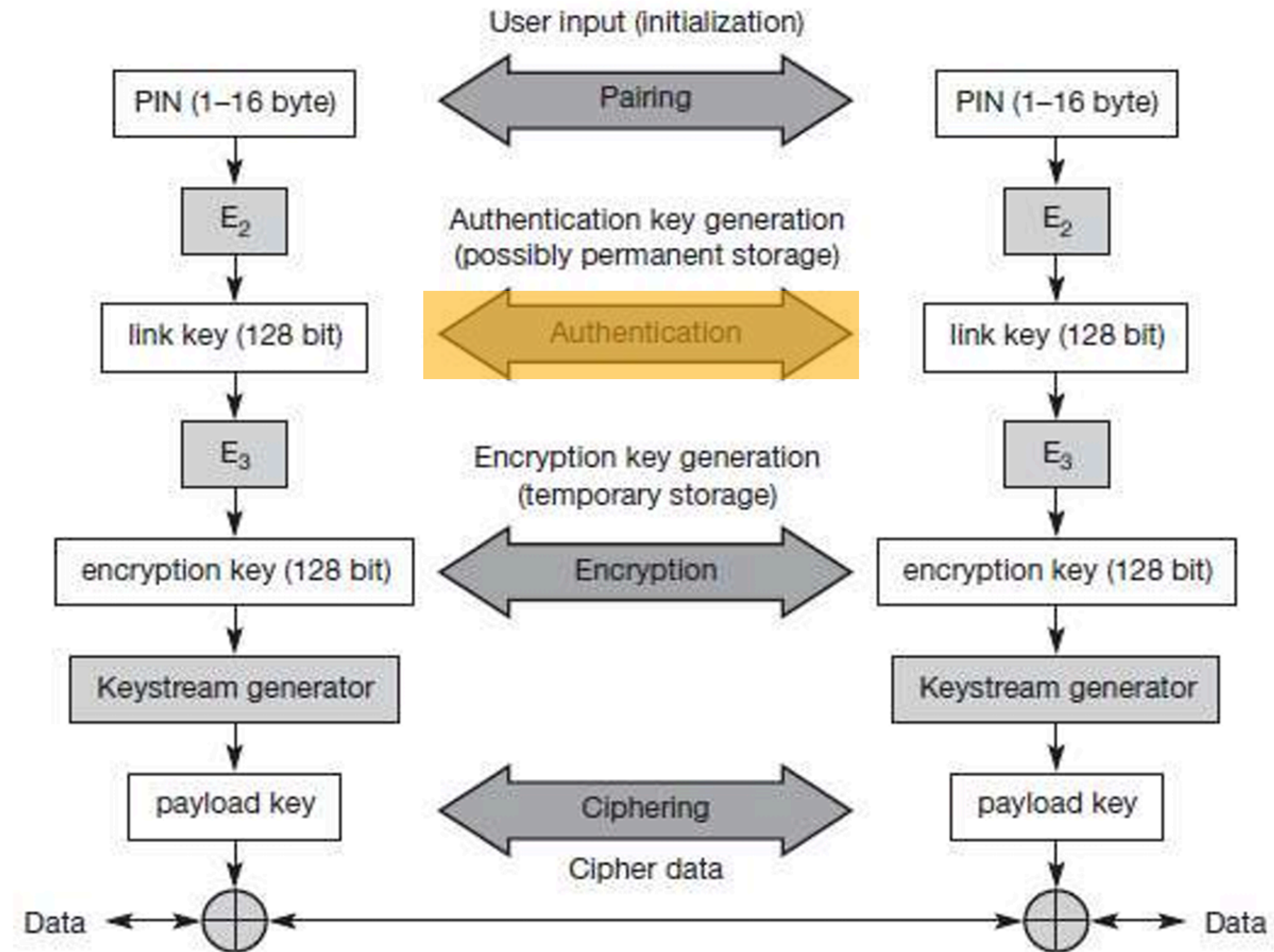
- The first step, called **pairing**, is necessary if two Bluetooth devices have never met before.
- To set up trust between the two devices a user can enter a secret PIN into both devices.
- This PIN can have a length of up to 16 byte.
- Based on the PIN, the device address, and random numbers, several keys can be computed which can be used as link key for authentication.



Bluetooth security architecture

Bluetooth Security

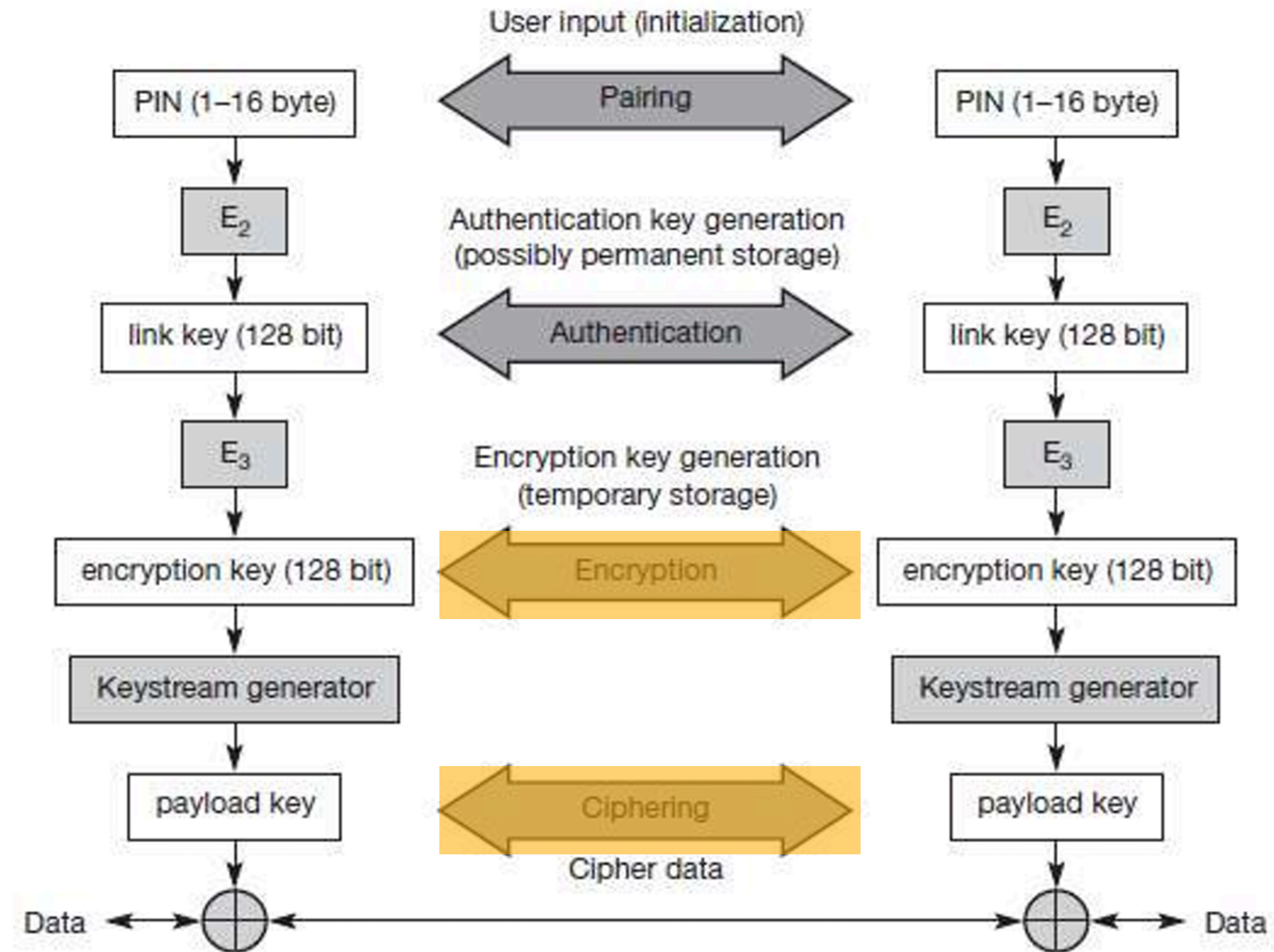
- The authentication is a **challenge-response** process.
- This process is based on the link key, a random number generated by a verifier (the device that requests authentication), and the device address of the claimant (the device that is authenticated).



Bluetooth security architecture

Bluetooth Security

- Based on the link key, and again a random number an encryption key is generated during the encryption stage of the security architecture.
- This key has a maximum size of 128 bits and can be individually generated for each transmission.
- Based on the encryption key, the device address and the current clock a **payload key** is generated for ciphering user data.
- The payload key is a stream of pseudo-random bits.
- The ciphering process is a simple XOR of the user data and the payload key.



Bluetooth security architecture

Questions?